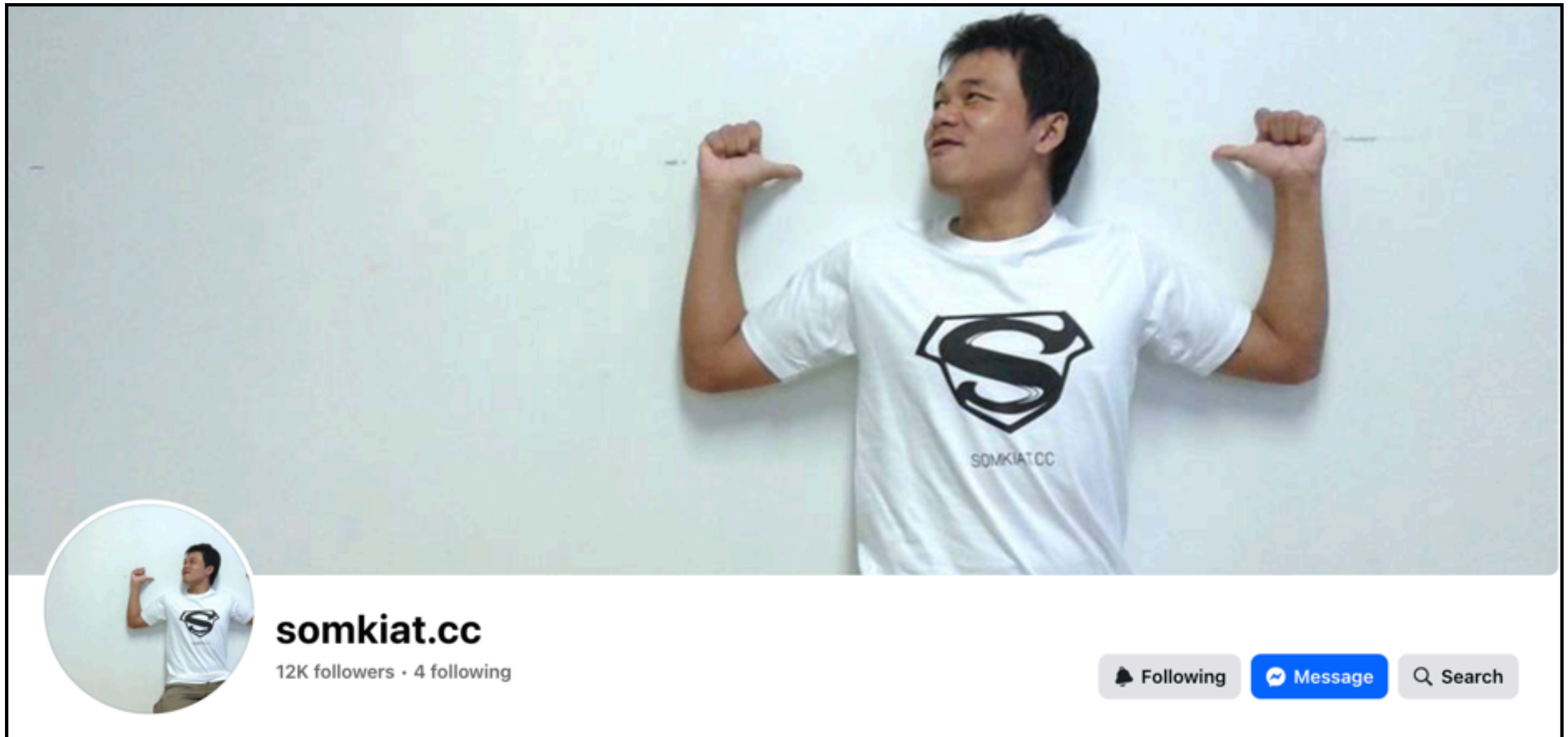




# AI Agent for Coding Code Smarter, Not Harder





## Software Developer at SCK



**Slide นี้ คนทำเองนะครับ  
ไม่ได้ใช้ Gen AI !!**



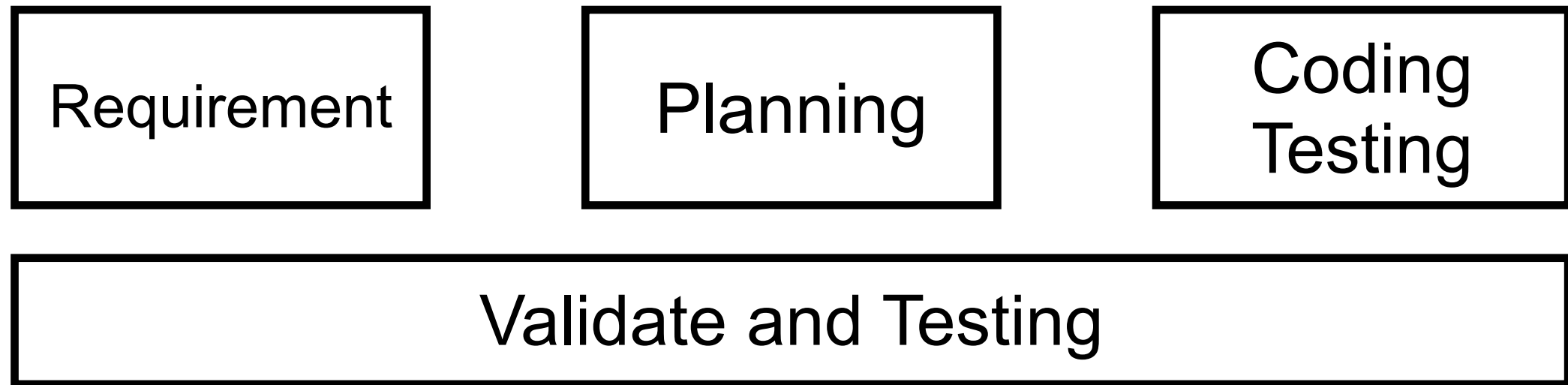
# **AI Agent for Coding**

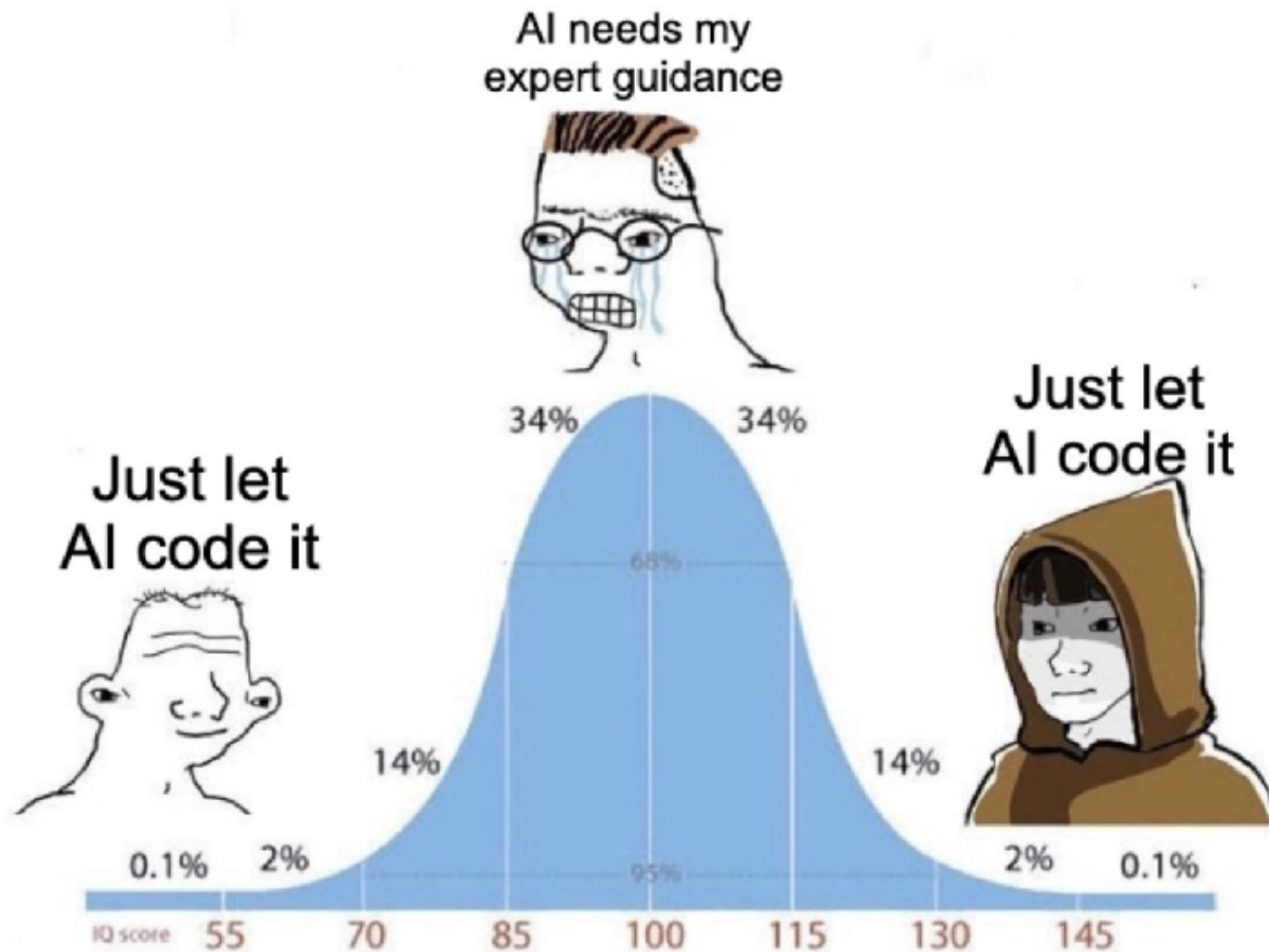
## **Code smarter, Not harder**



# Topics

Software development workflows with AI Agent  
Spec-Driven Development (SDD)  
Demo coding with AI Agent





# Programming/Coding





# History of Programming

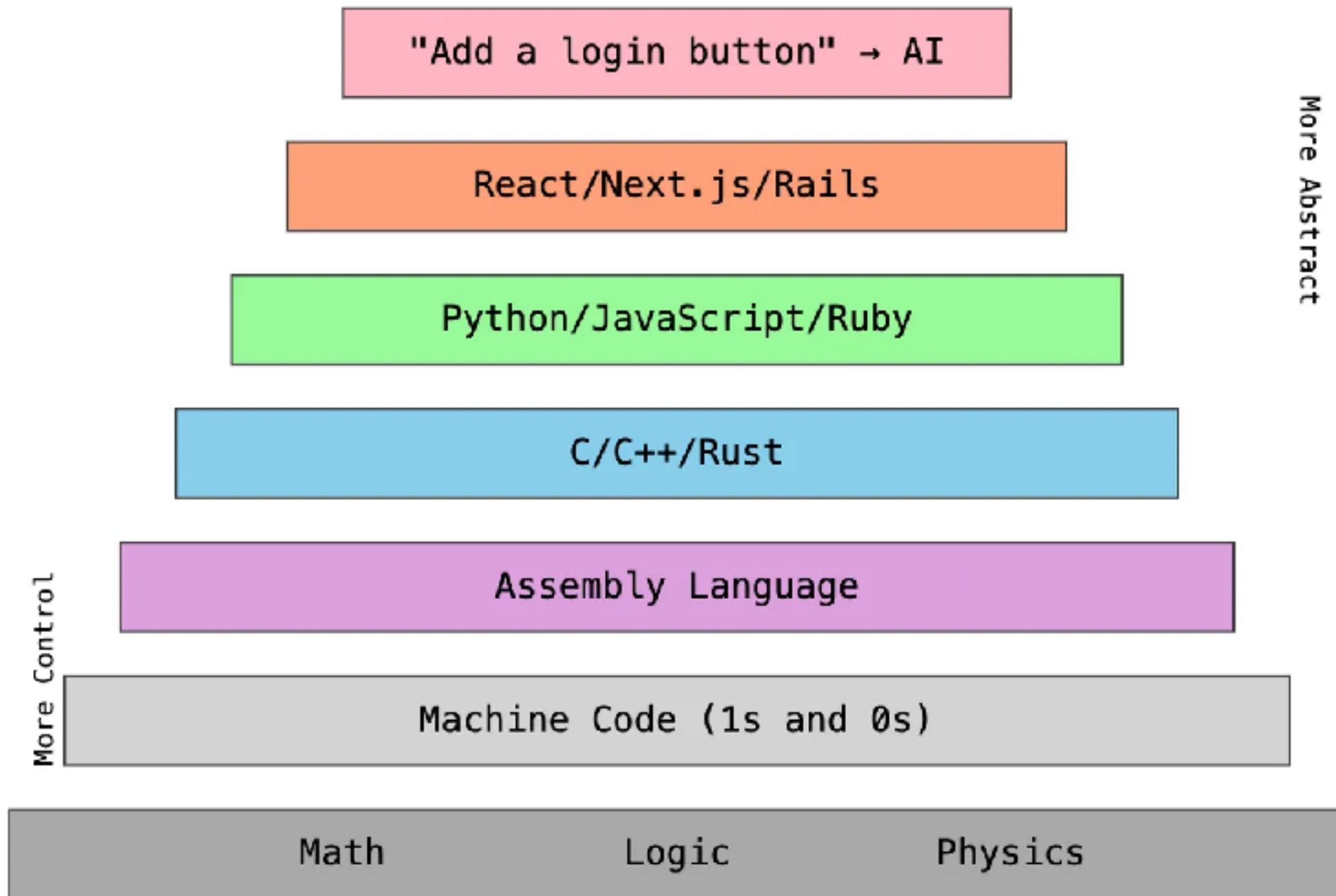


Increase of abstraction  
Productivity  
Knowledge gap !!





# The Modern Tower of Abstraction



<https://cline.bot/blog/from-assembly-to-ai-why-vibe-coding-is-just-another-chapter-in-our-abstraction-story>





Andrej Karpathy ✓

@karpathy



There's a new kind of coding I call "vibe coding", where you fully give in to the vibes, embrace exponentials, and forget that the code even exists. It's possible because the LLMs (e.g. Cursor Composer w Sonnet) are getting too good. Also I just talk to Composer with SuperWhisper so I barely even touch the keyboard. I ask for the dumbest things like "decrease the padding on the sidebar by half" because I'm too lazy to find it. I "Accept All" always, I don't read the diffs anymore. When I get error messages I just copy paste them in with no comment, usually that fixes it. The code grows beyond my usual comprehension, I'd have to really read through it for a while. Sometimes the LLMs can't fix a bug so I just work around it or ask for random changes until it goes away. It's not too bad for throwaway weekend projects, but still quite amusing. I'm building a project or webapp, but it's not really coding - I just see stuff, say stuff, run stuff, and copy paste stuff, and it mostly works.

6:17 AM · Feb 3, 2025 · 5.3M Views

<https://x.com/karpathy/status/1886192184808149383>



# Vibe Coding

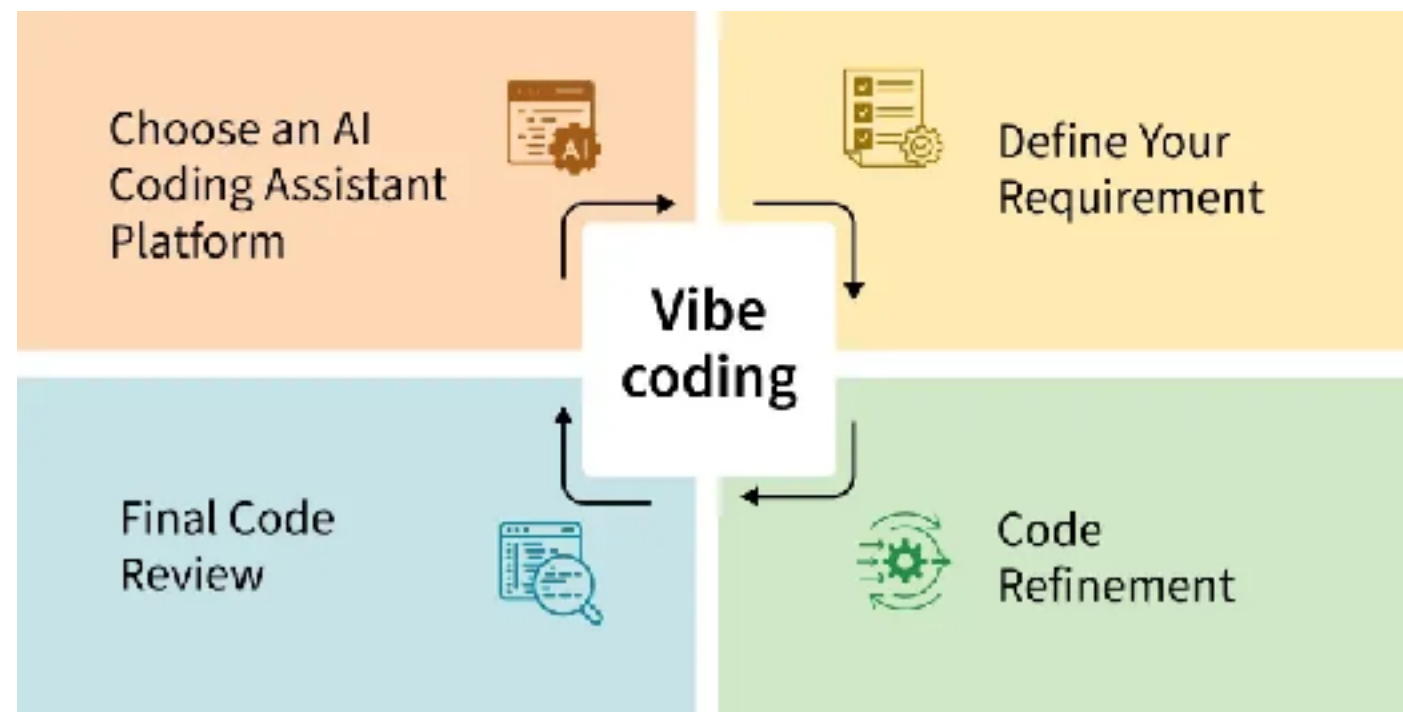
**Software development practice** that uses artificial intelligence (AI) to generate functional code from **natural language prompts**, accelerating development, and making app building more accessible, especially for those with limited programming experience

<https://cloud.google.com/discover/what-is-vibe-coding>



# Types of Vibe coding ?

Pure vibe coding  
**Responsible AI-assist development**



<https://cloud.google.com/discover/what-is-vibe-coding>



# **Working with AI Agent !!**



**"codex is better"**

**"claude.md"**

**"i just vibecoded"**

**"skills"**

**"ralph"**

**"agentic"**

**"obsidian"**

**"mcp"**

**"mac mini"**

**"clawdbot"**

**"agent sdk"**

**"remotion"**

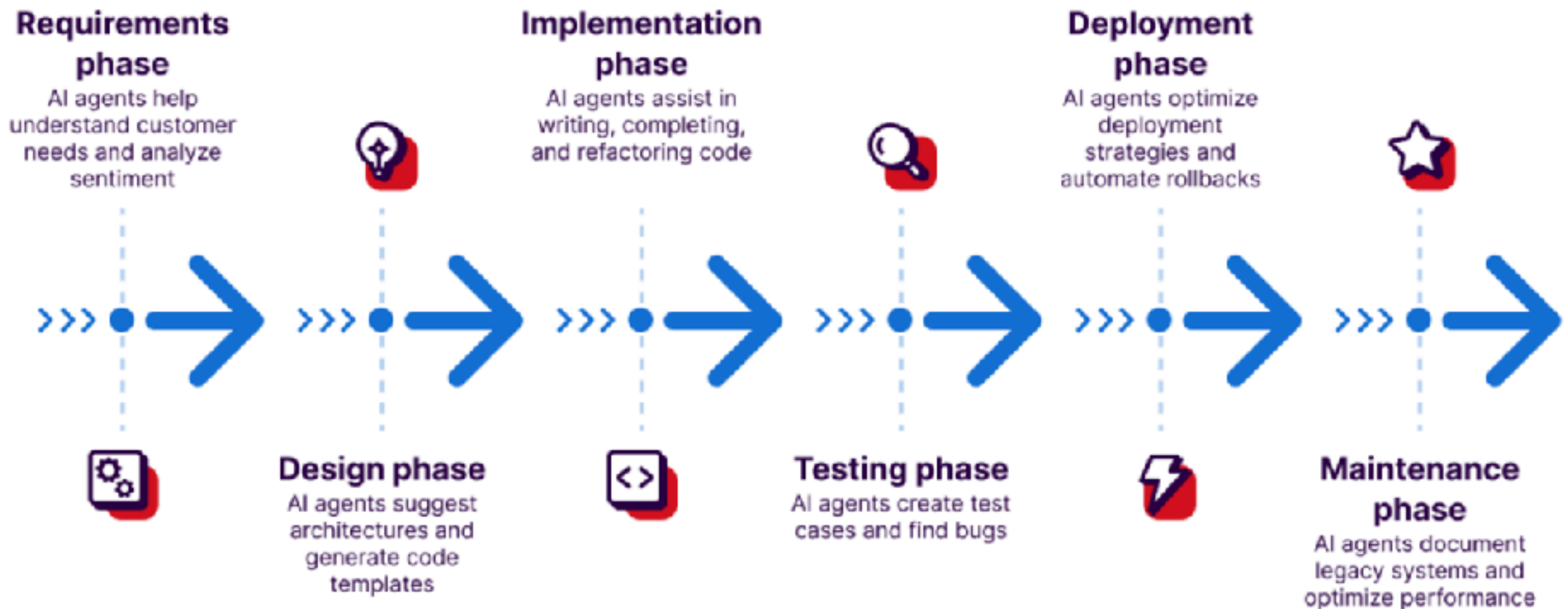




# **AI is the new normal in software development**

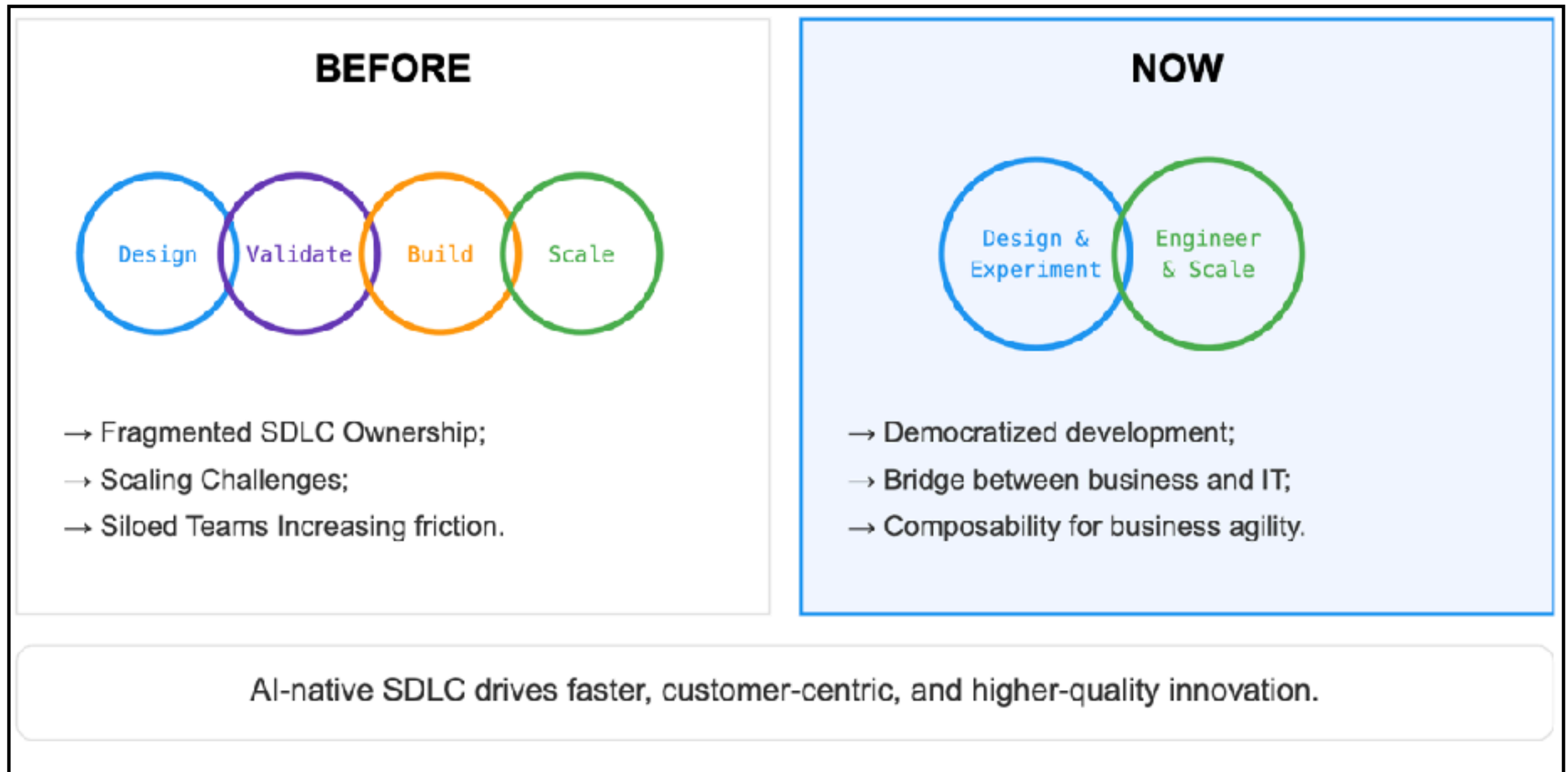


# AI in Software Development

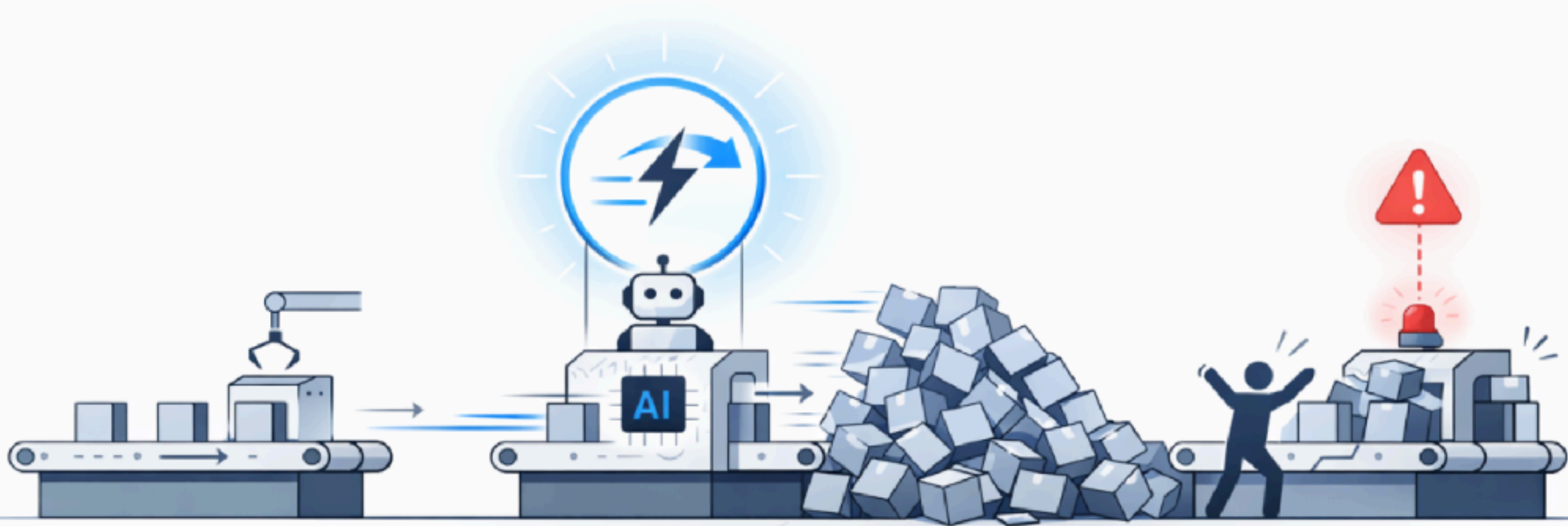




# AI in Software Development

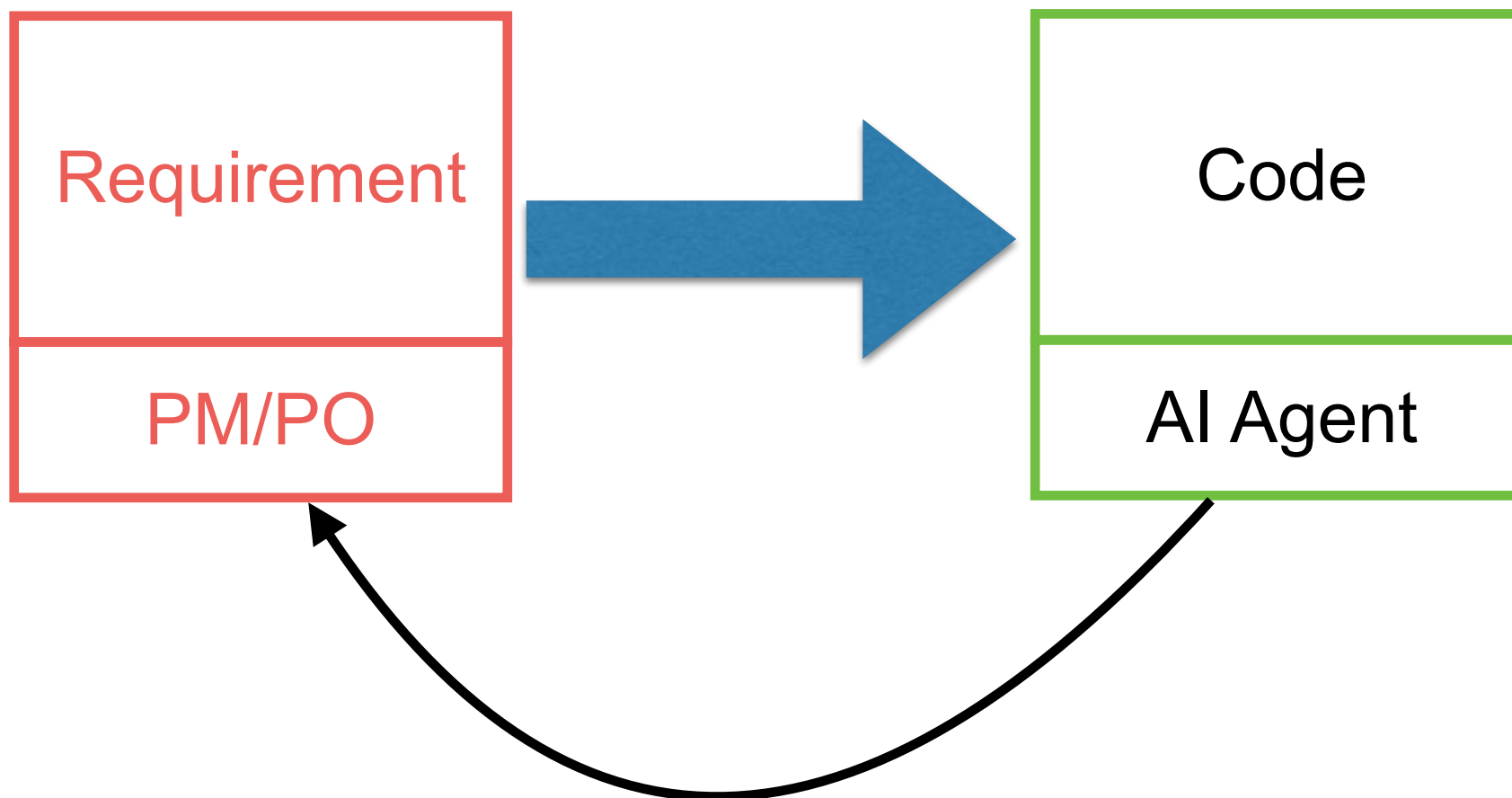


# Invest in Real Bottleneck ?



# New Bottleneck ?

Decide what to **build**, not code



**Get feedback faster**

<https://www.theaiopportunities.com/p/is-ai-slowng-down-the-new-bottleneck>



# DORA



## State of AI-assisted Software Development

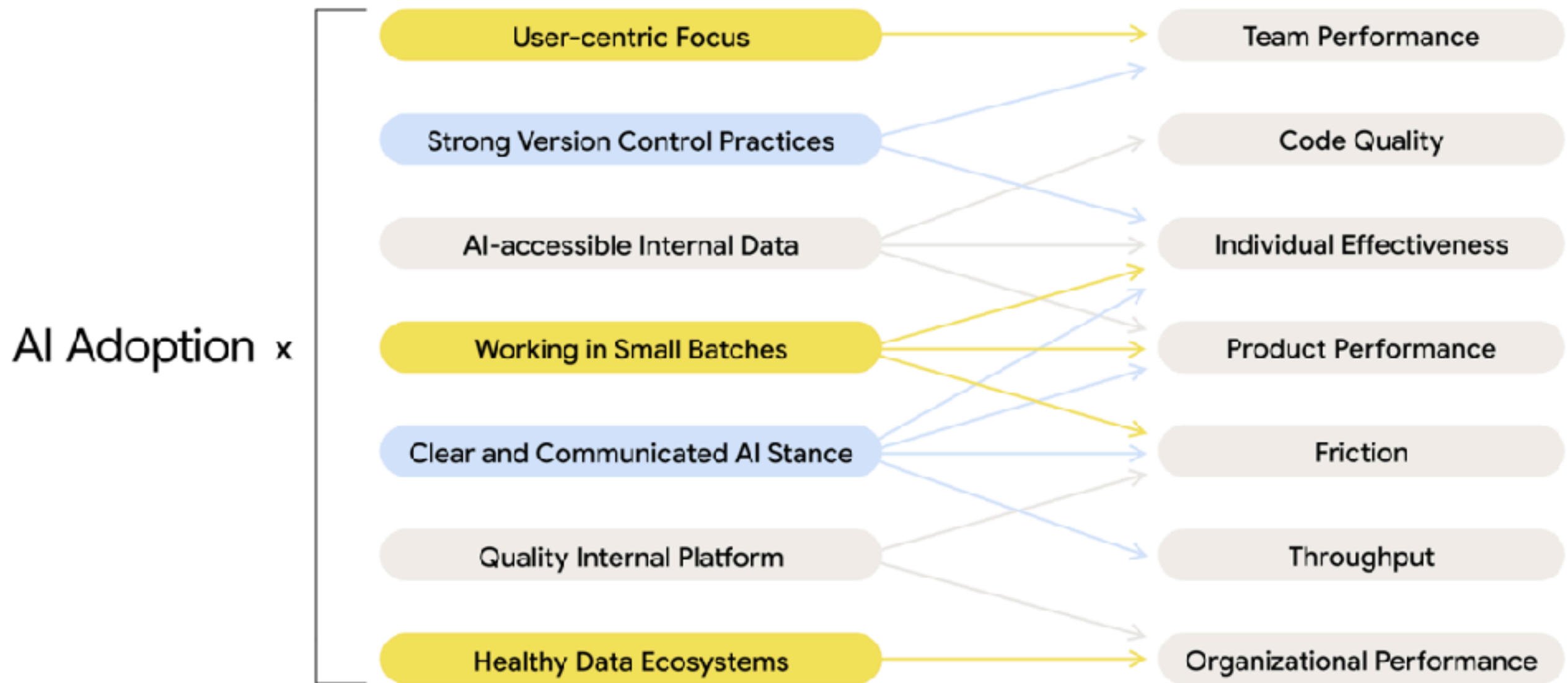
Google Cloud

2025

<https://cloud.google.com/blog/products/ai-machine-learning/announcing-the-2025-dora-report>



# DORA AI Capabilities Model



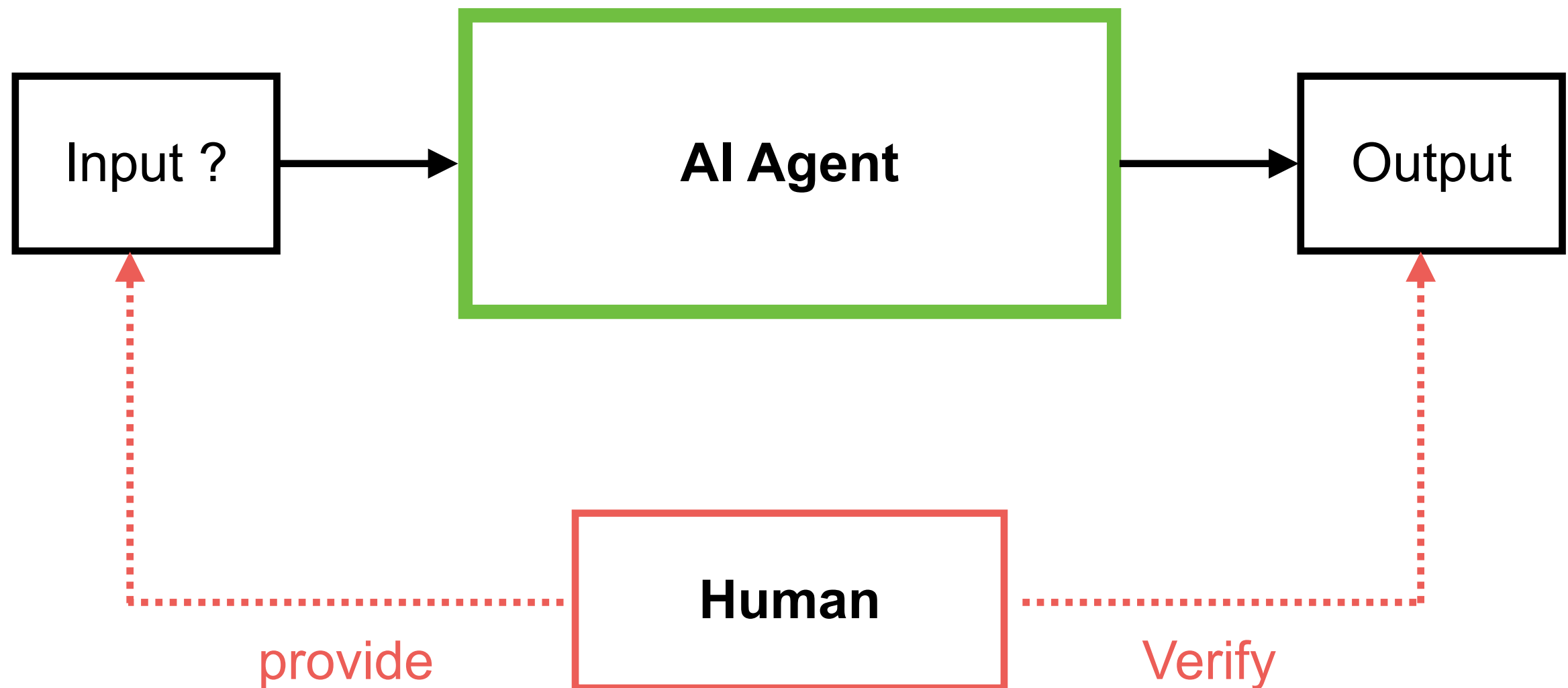
<https://cloud.google.com/blog/products/ai-machine-learning/announcing-the-2025-dora-report>



# AI Agent for Coding



# AI Agent + Human in the Loop



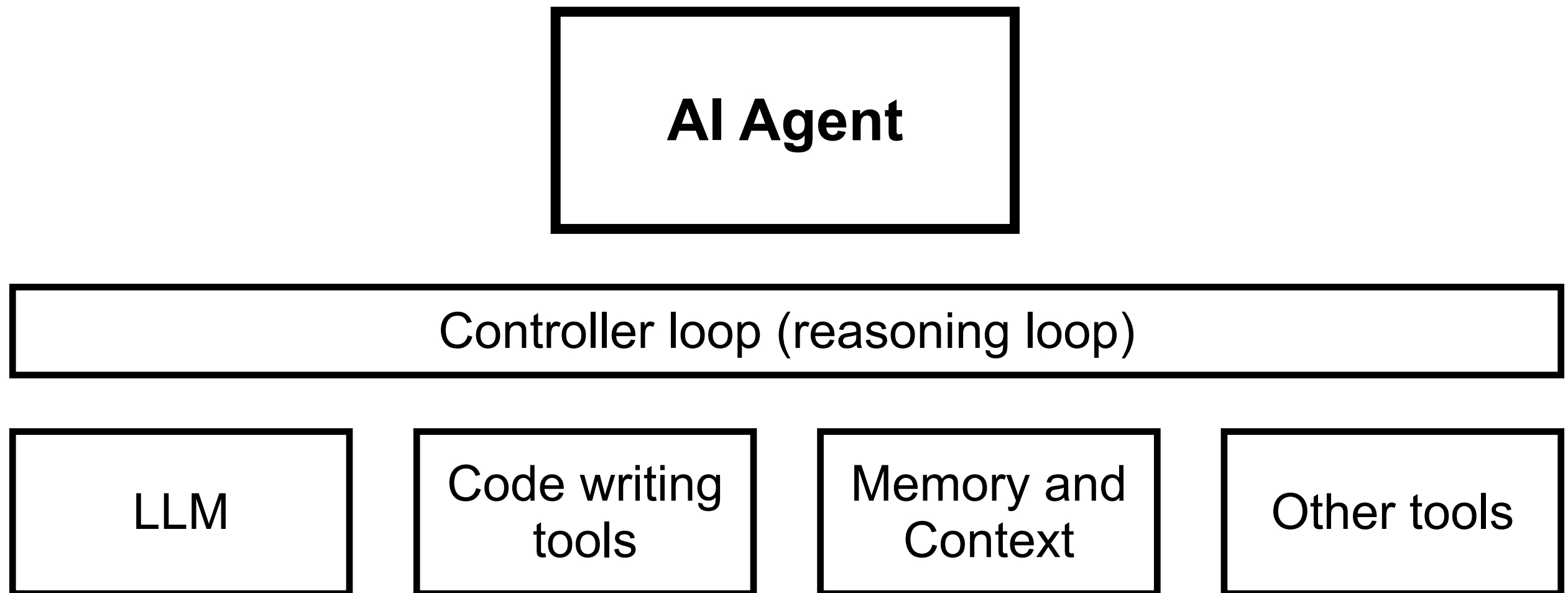


# **AI draft**

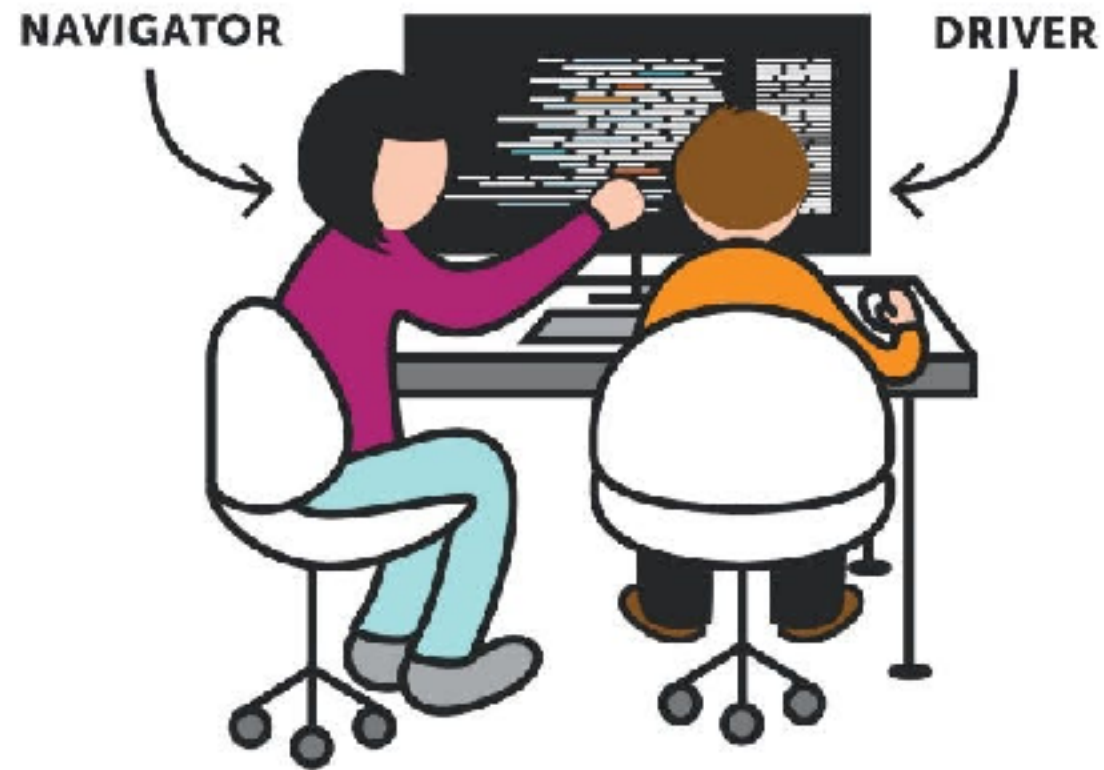
# **Human craft**



# AI Agent for Coding



## PAIR PROGRAMMING

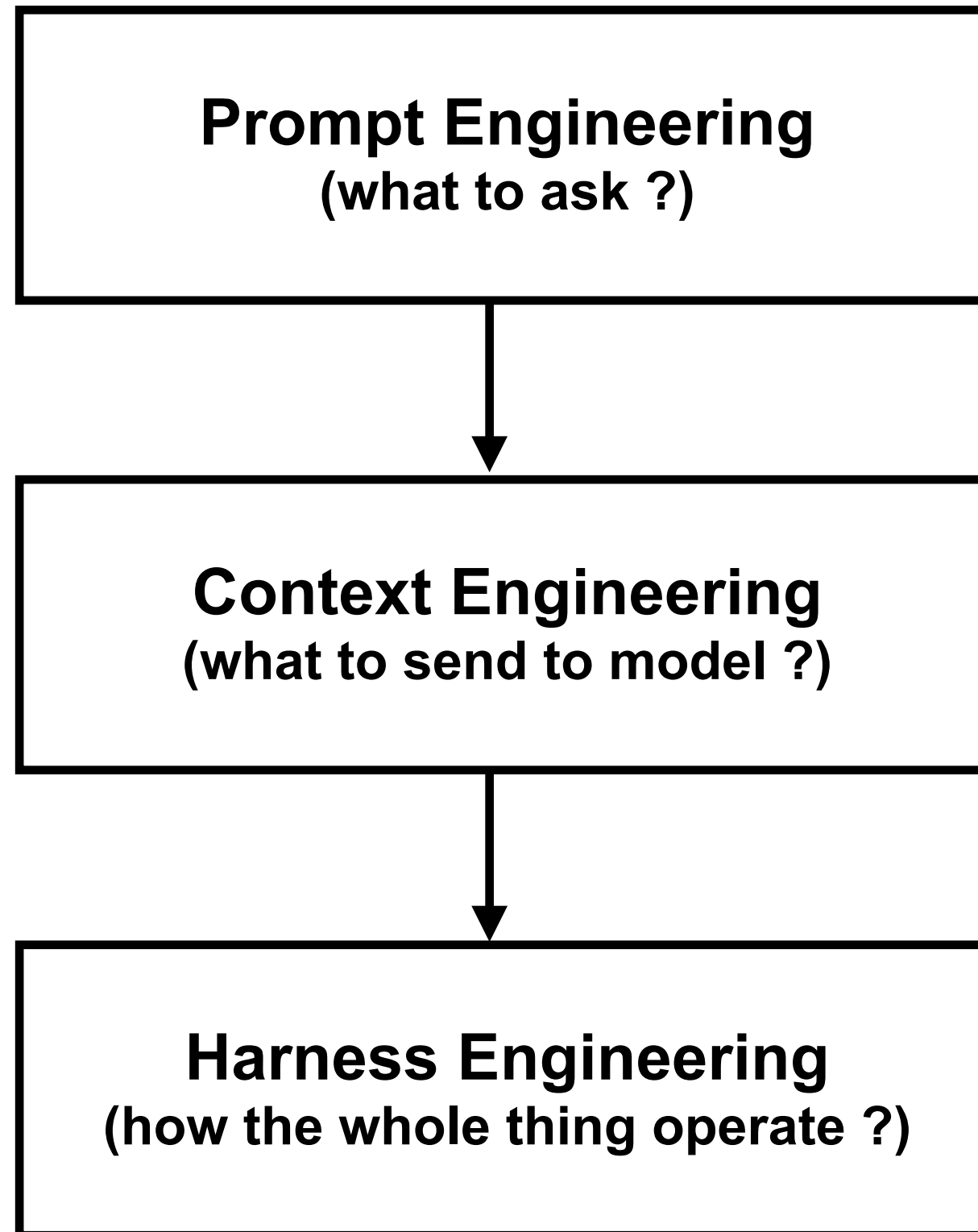


Reviews, Tests and Understands  
with experience



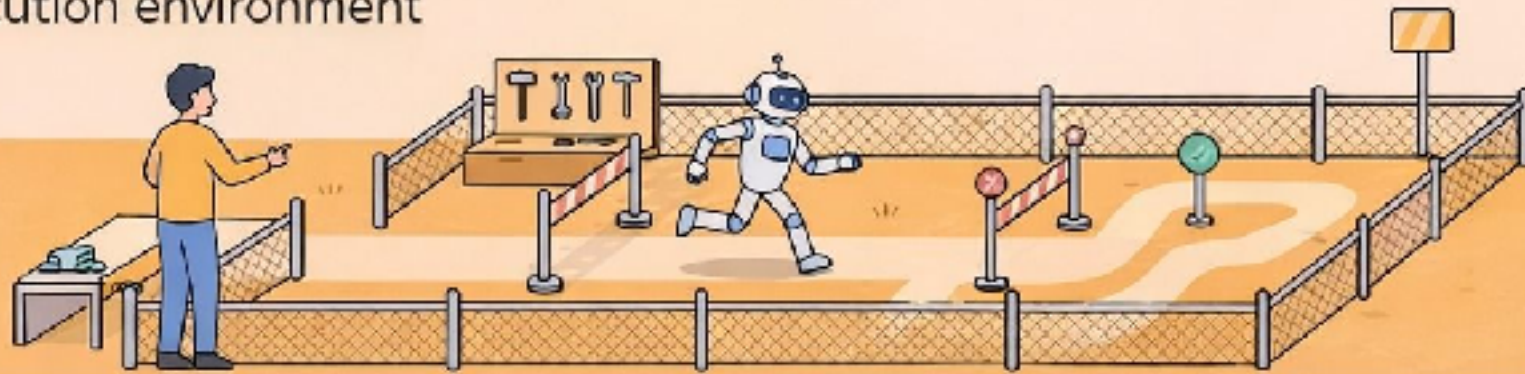
# How to work with AI Agent ?





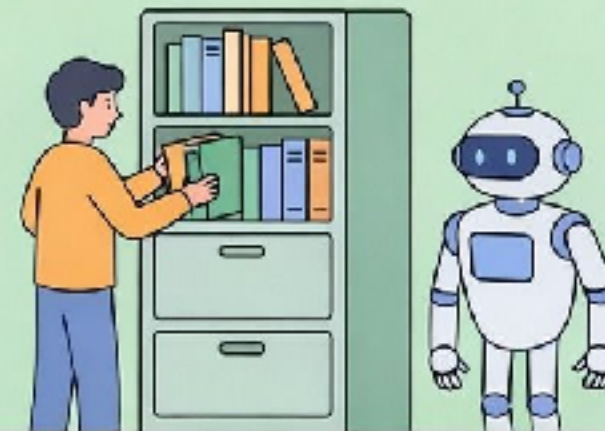
## Harness Engineering — Build the World

Design the execution environment



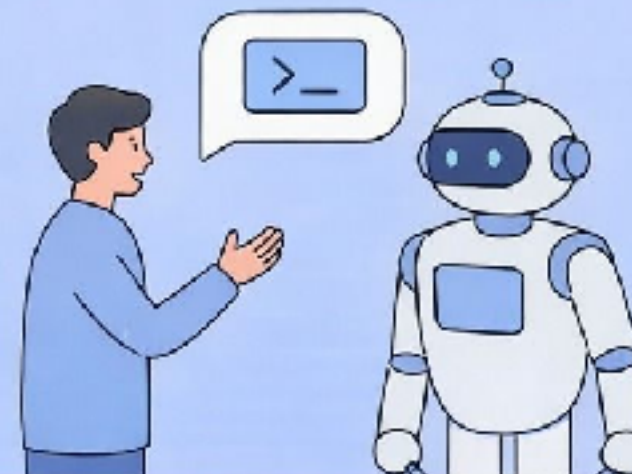
## Context Engineering — What to See

Manage visible information



## Prompt Engineering — What to Say

Optimize single-turn interactions

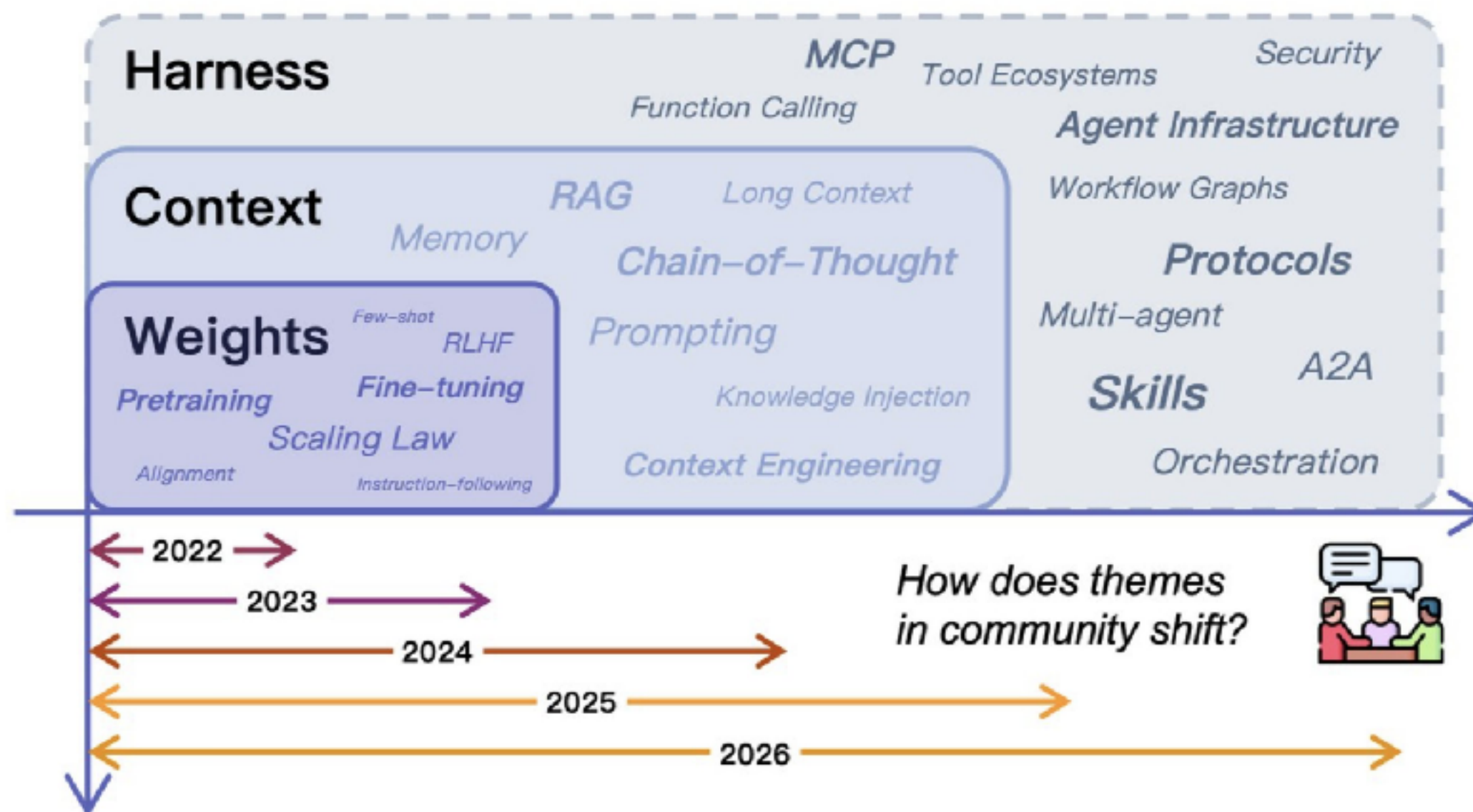


<https://milvus.io/blog/harness-engineering-ai-agents.md>





## 2 Background: From Weights to Context to Harness

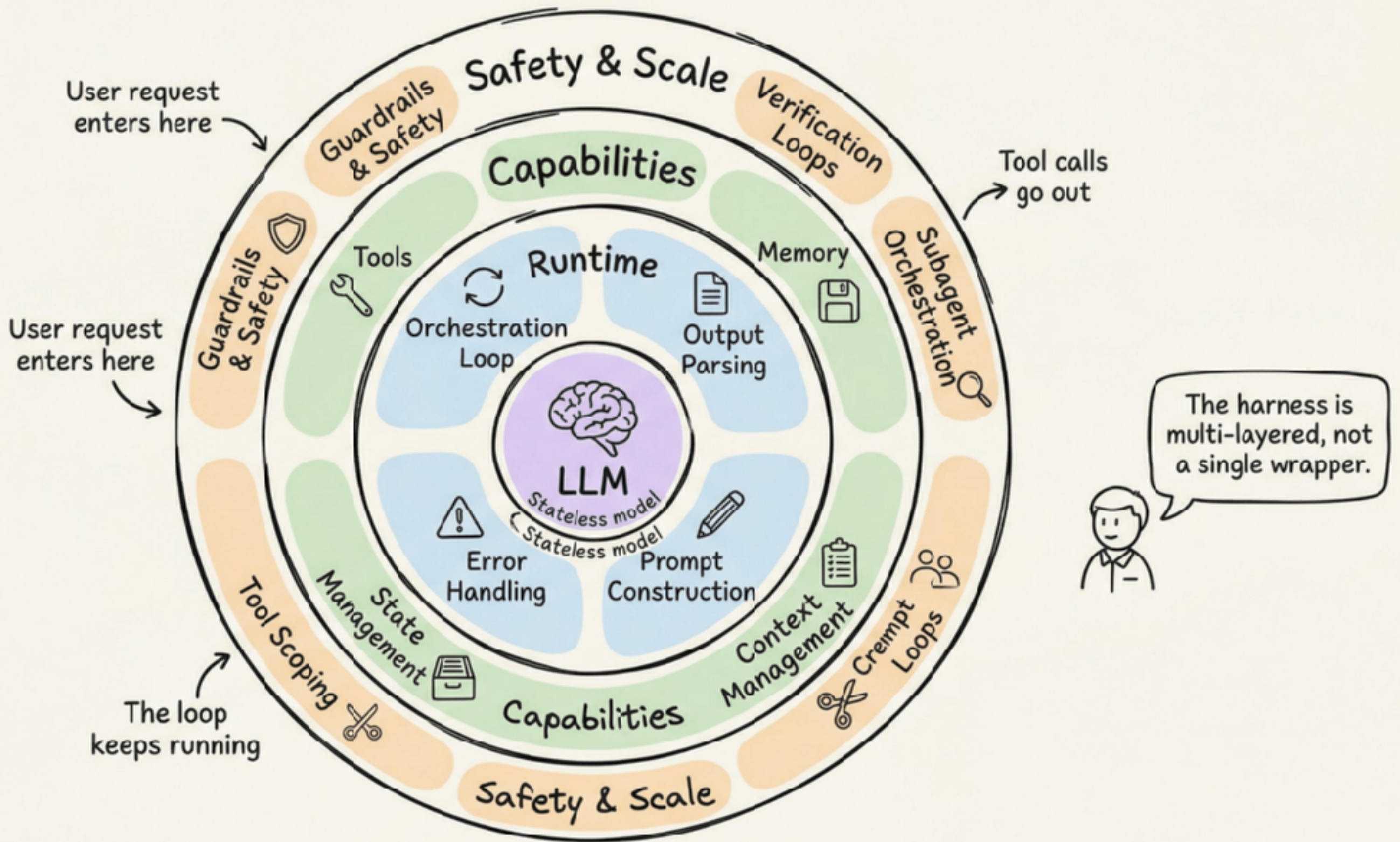


**Figure 2 Community theme evolution across three capability layers.** The stacked layers—Weights, Context, and Harness—show how the center of gravity in the LLM agent community has shifted outward over time, from parametric knowledge and prompting toward harness-level infrastructure such as tool ecosystems, protocols, skills, and multi-agent orchestration.

<https://arxiv.org/pdf/2604.08224>







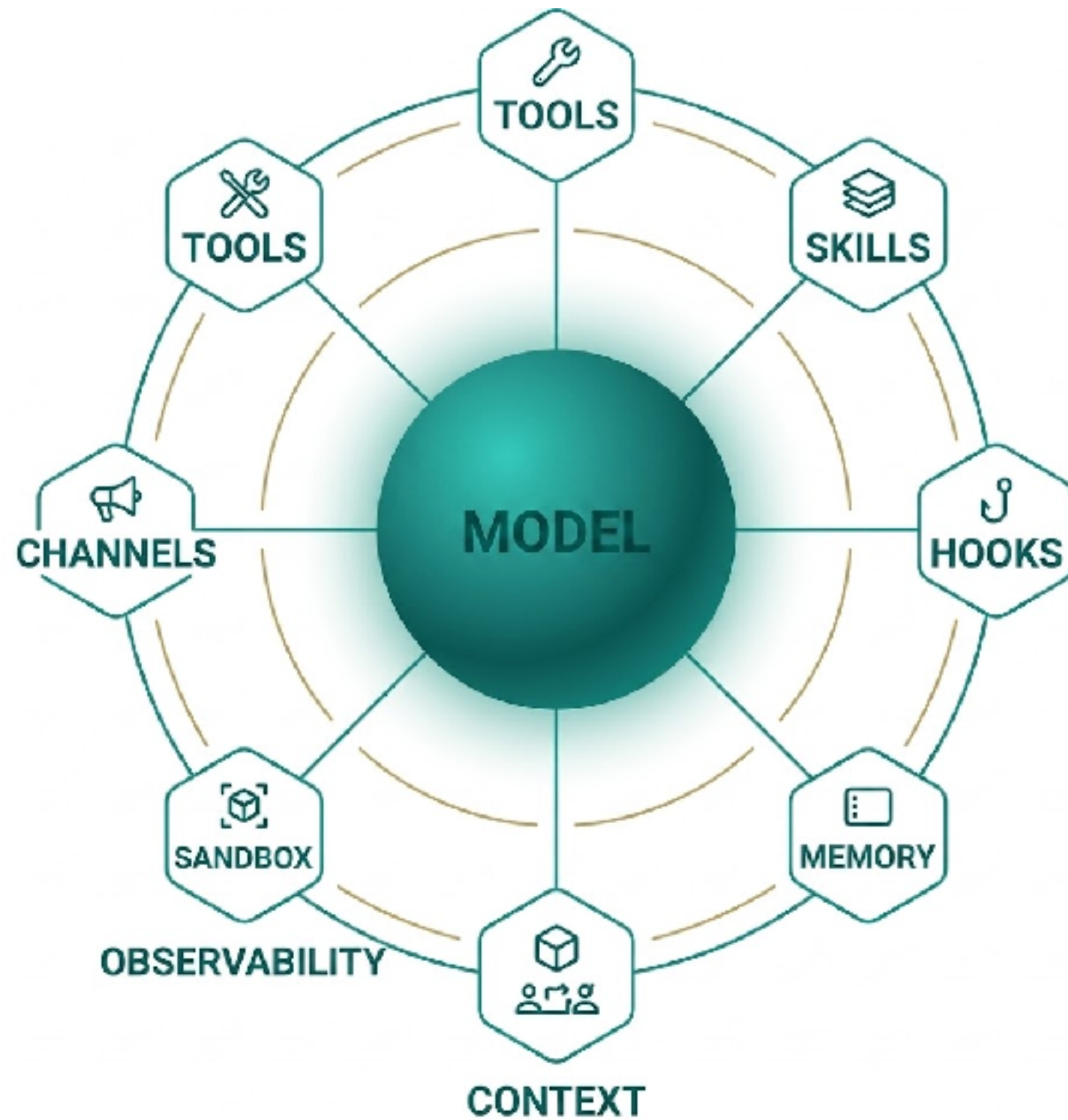
<https://blog.dailydoseofds.com/p/evolution-of-agent-landscape-from>



**AI Agent = LLM + Harness**

**Harness = AI Agent - LLM**





# Workflows with AI Agent ?



# Workflows ?

Application level

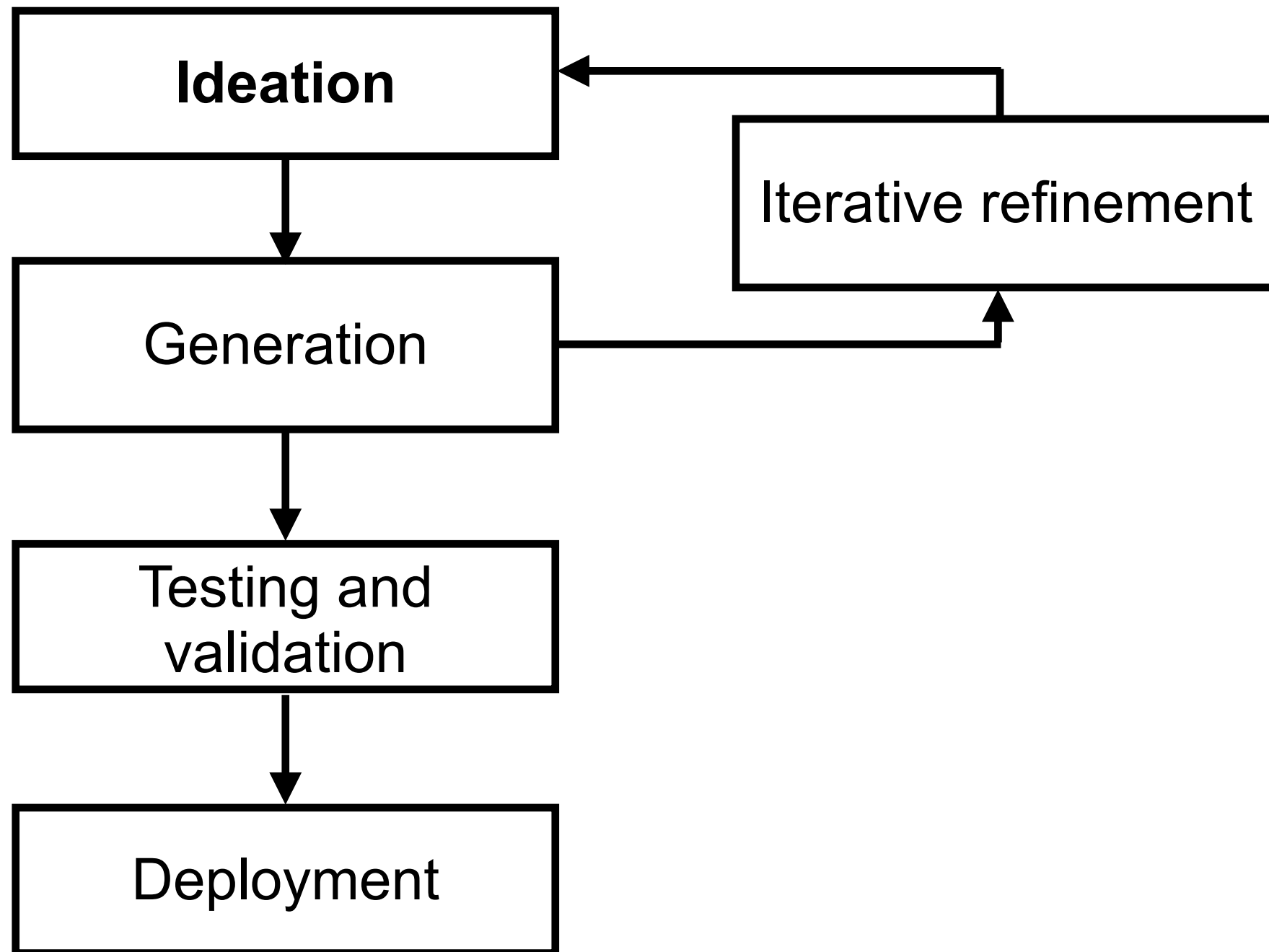
Code level

<https://cloud.google.com/discover/what-is-vibe-coding>



# Application level workflow

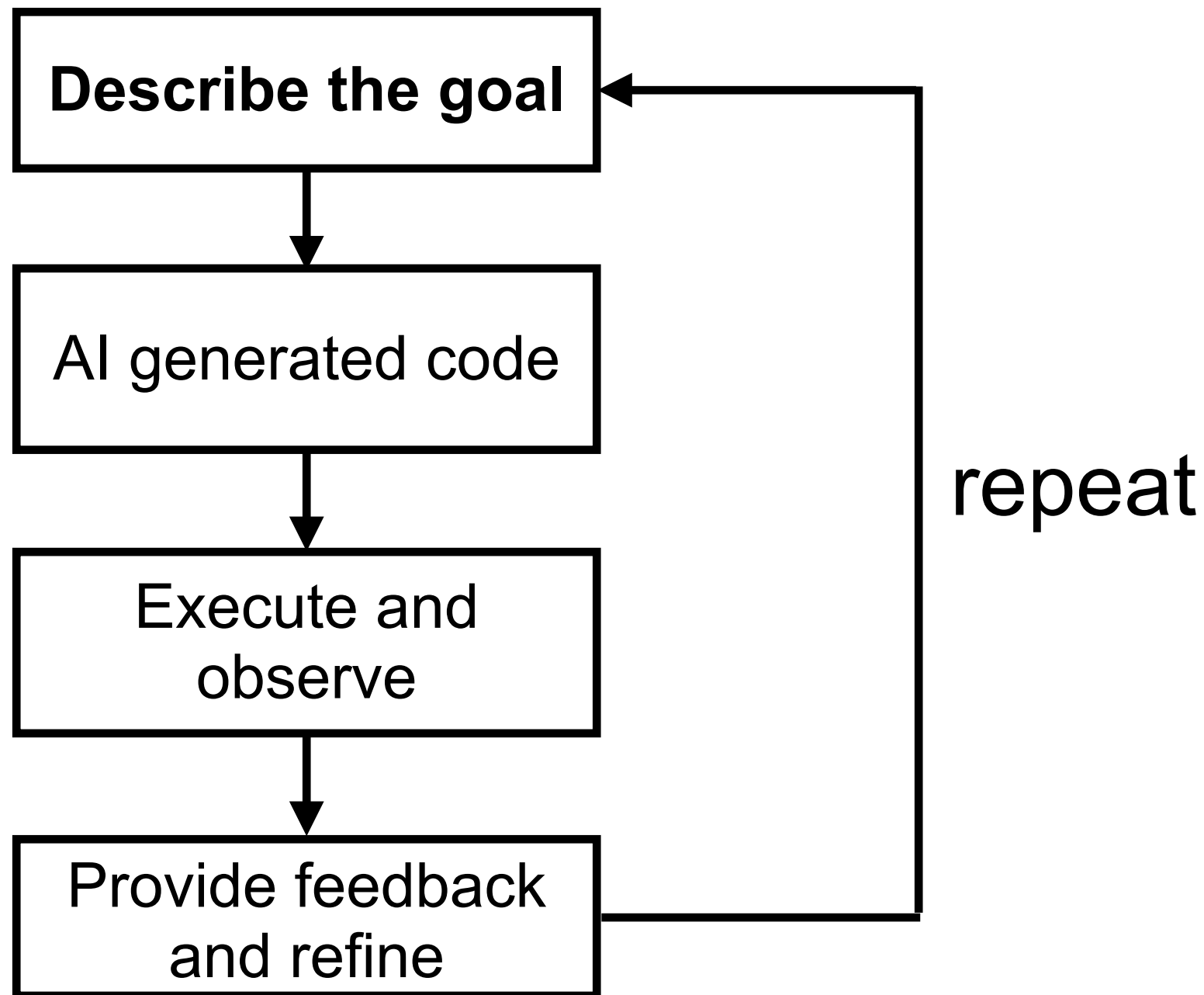
Talking about **high-level idea** from concept to deployment





# Code level workflow

Conversational loop to create a **specific code**



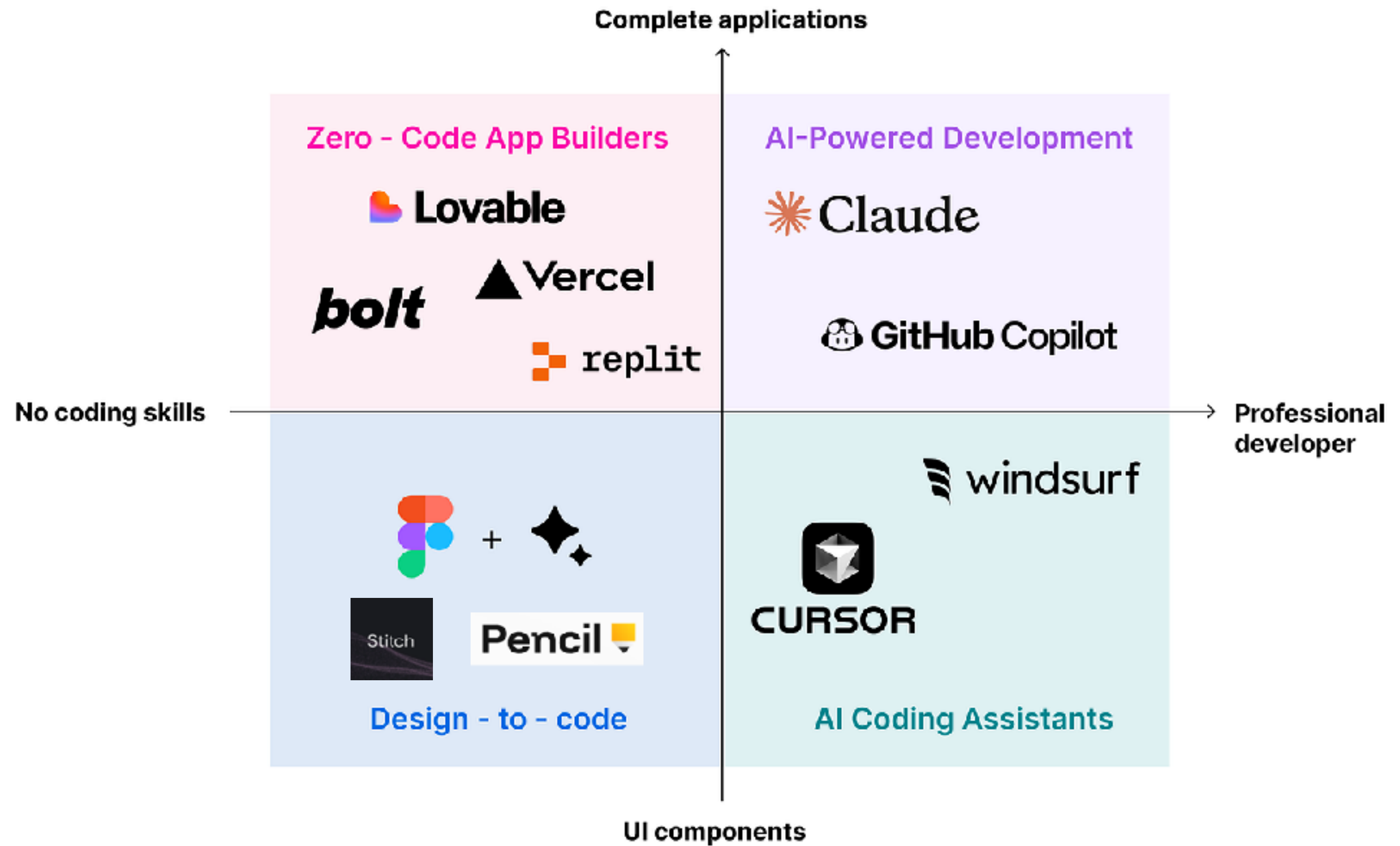


# Tools ?



# AI Agent for Coding !!

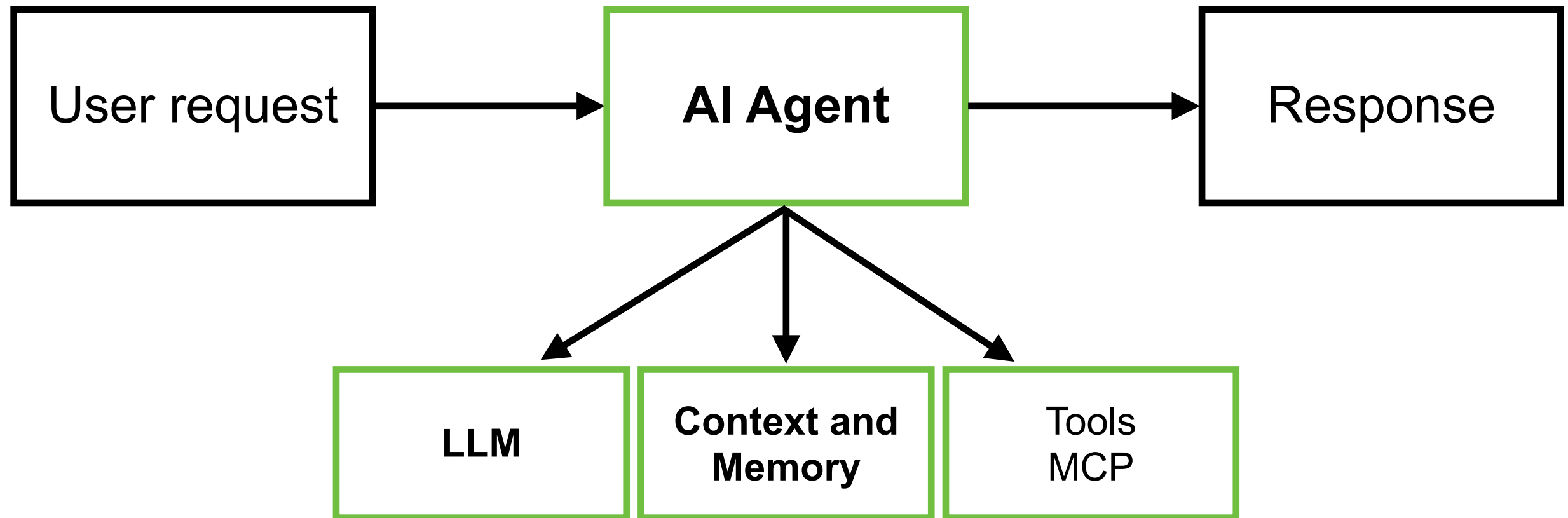




# Before start ...



# AI Agent for Coding



**How to manage context or knowledge for Agent ?**



**Token is new oil !!**  
**Token is expensive !!**



# LLM Model :: GPT 5.5

**GPT-5.5** Default Compare Try in Playground

A new class of intelligence for coding and professional work.

**REASONING**  
  
Highest

**SPEED**  
  
Fast

**PRICE**  
\$5 - \$30  
Input - Output

**INPUT**  
     
Text, image

**OUTPUT**  
     
Text

GPT-5.5 is our newest frontier model for the most complex professional work. Learn more in our [latest model guide](#). Reasoning effort supports: none, low, medium (default), high and xhigh.

- ✦ 1,050,000 context window
- ➡ 128,000 max output tokens
- 📅 Dec 01, 2025 knowledge cutoff
- 🗨 Reasoning token support

---

**Pricing**

Pricing is based on the number of tokens used, or other metrics based on the model type. For tool-specific models, like search and computer use, there's a fee per tool call. See details in the [pricing page](#).

Text tokens Per 1M tokens - Batch API price ☐

|                        |                               |                          |
|------------------------|-------------------------------|--------------------------|
| Input<br><b>\$5.00</b> | Cached input<br><b>\$0.50</b> | Output<br><b>\$30.00</b> |
|------------------------|-------------------------------|--------------------------|

<https://developers.openai.com/api/docs/models/gpt-5.5>





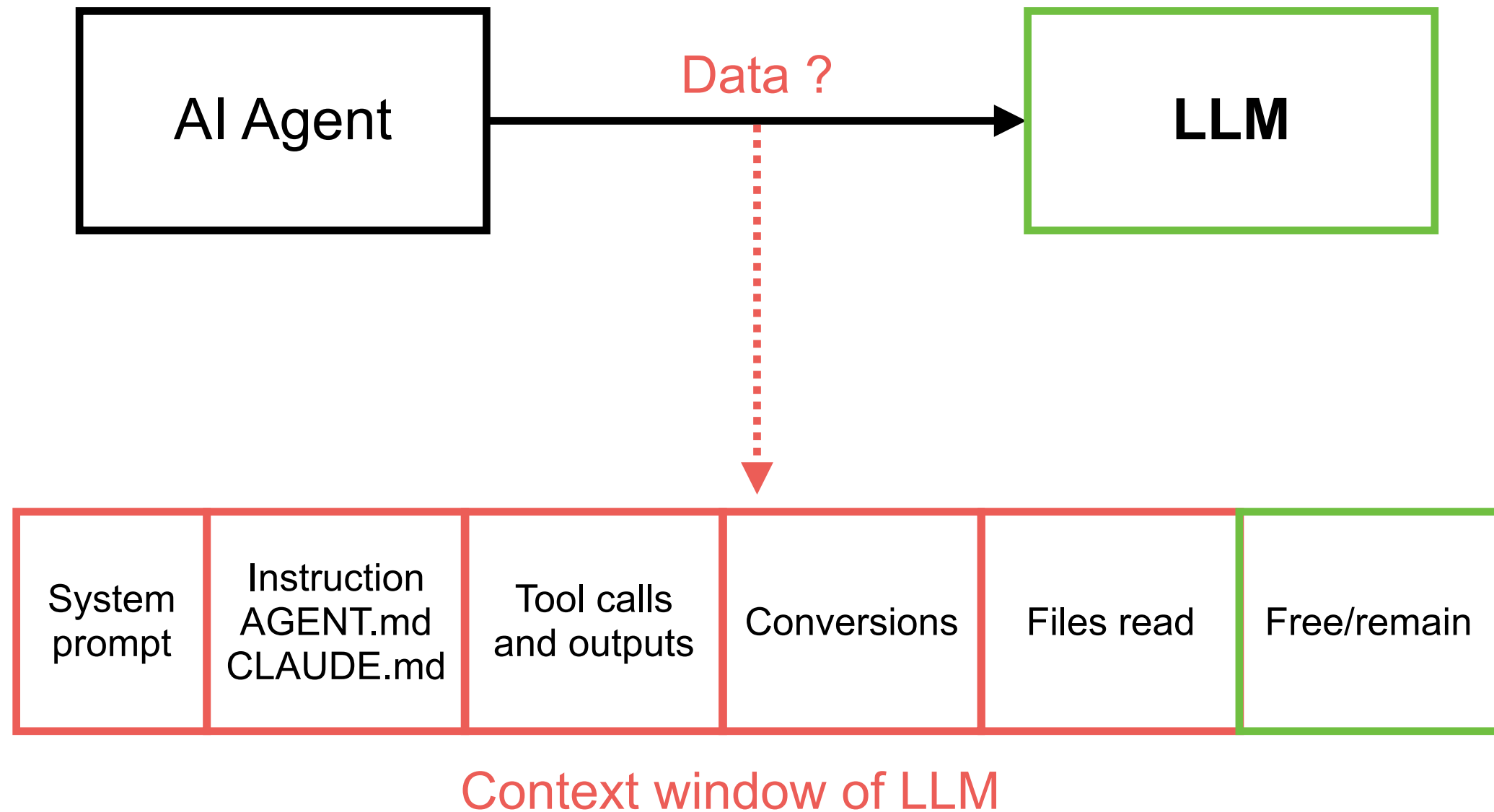
# LLM Model :: Anthropic

| Model             | Base Input Tokens | 5m Cache Writes | 1h Cache Writes | Cache Hits & Refreshes | Output Tokens |
|-------------------|-------------------|-----------------|-----------------|------------------------|---------------|
| Claude Opus 4.7   | \$5 / MTok        | \$6.25 / MTok   | \$10 / MTok     | \$0.50 / MTok          | \$25 / MTok   |
| Claude Opus 4.6   | \$5 / MTok        | \$6.25 / MTok   | \$10 / MTok     | \$0.50 / MTok          | \$25 / MTok   |
| Claude Opus 4.5   | \$5 / MTok        | \$6.25 / MTok   | \$10 / MTok     | \$0.50 / MTok          | \$25 / MTok   |
| Claude Opus 4.1   | \$15 / MTok       | \$18.75 / MTok  | \$30 / MTok     | \$1.50 / MTok          | \$75 / MTok   |
| Claude Opus 4     | \$15 / MTok       | \$18.75 / MTok  | \$30 / MTok     | \$1.50 / MTok          | \$75 / MTok   |
| Claude Sonnet 4.6 | \$3 / MTok        | \$3.75 / MTok   | \$6 / MTok      | \$0.30 / MTok          | \$15 / MTok   |
| Claude Sonnet 4.5 | \$3 / MTok        | \$3.75 / MTok   | \$6 / MTok      | \$0.30 / MTok          | \$15 / MTok   |
| Claude Sonnet 4   | \$3 / MTok        | \$3.75 / MTok   | \$6 / MTok      | \$0.30 / MTok          | \$15 / MTok   |

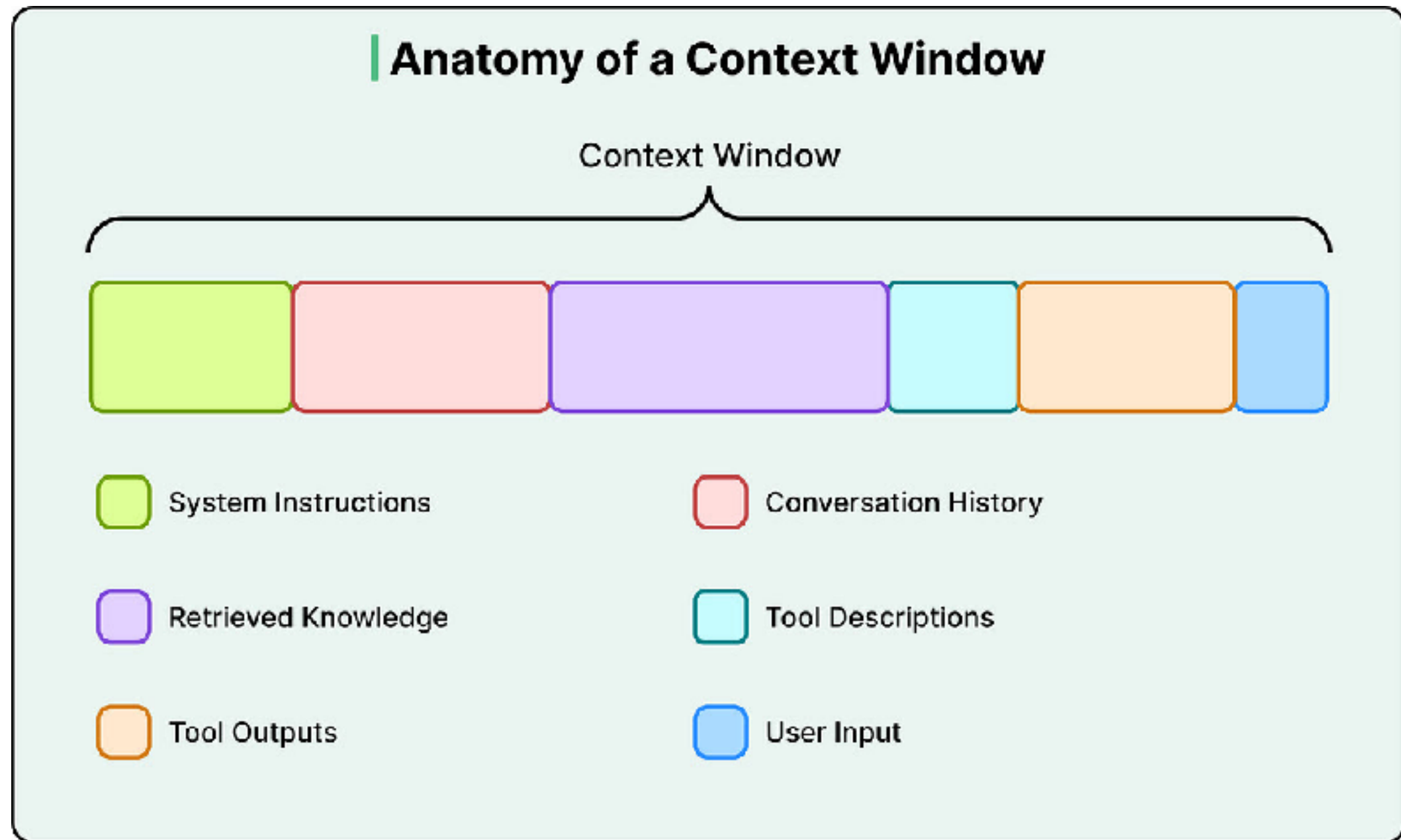
<https://platform.claude.com/docs/en/about-claude/pricing>



# Context Window ?



# Anatomy of Context Window



# Types of Knowledge/Context ?

## Shared context

Static Knowledge

Prompt engineering  
Global Instructions  
AGENTS.md  
CLUDE.md  
Rules

## Task context

Dynamic Knowledge

Memory  
Graph/Vector database  
MCP tools  
Specific Instructions  
**Agent SKILLS**

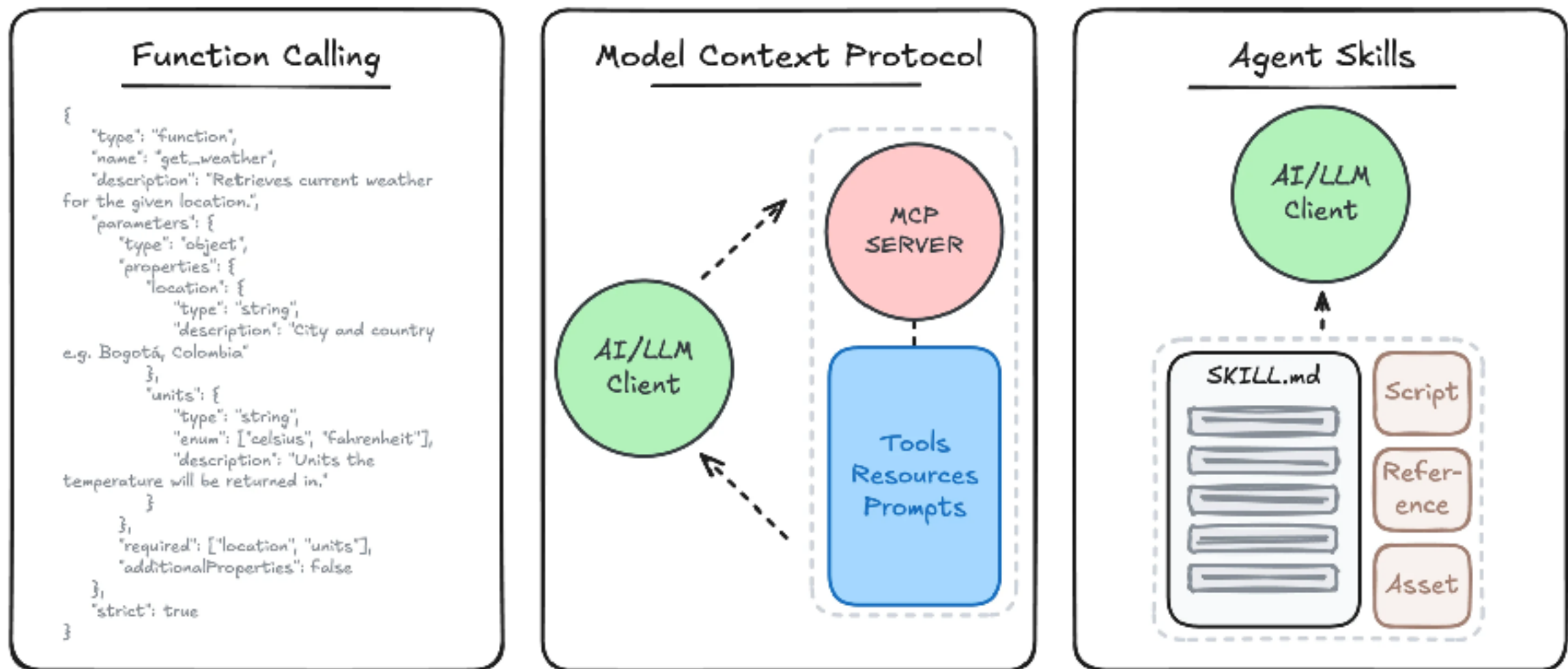
Sub-agent and Modes

**Context engineering ?**



# Agent Skills

Standard way to give AI agent new capabilities and expertise



<https://agentskills.io/home>



# How to manage context ?



# Context Engineering

Right information

Right tools

Structured context





# Long context == Better result ?



# Context window size of models

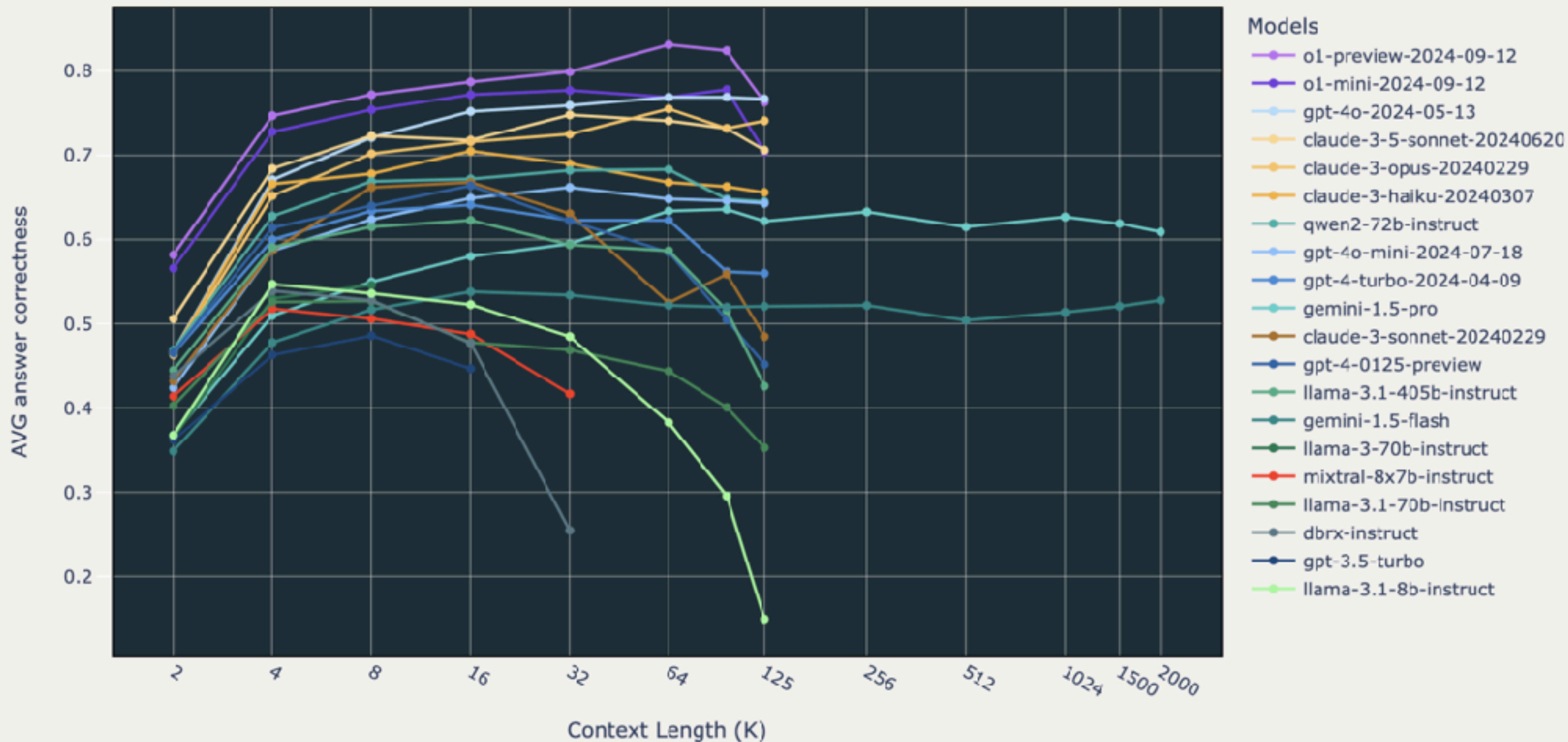
Different models may have different window size

| Model name      | Context window size |
|-----------------|---------------------|
| GPT 5.5         | 1M                  |
| Gemini 1.5, 2.5 | 1M                  |
| Llama 4         | 1M                  |
| Claude Opus 4.7 | 1M                  |



# Long context LLM is better !!

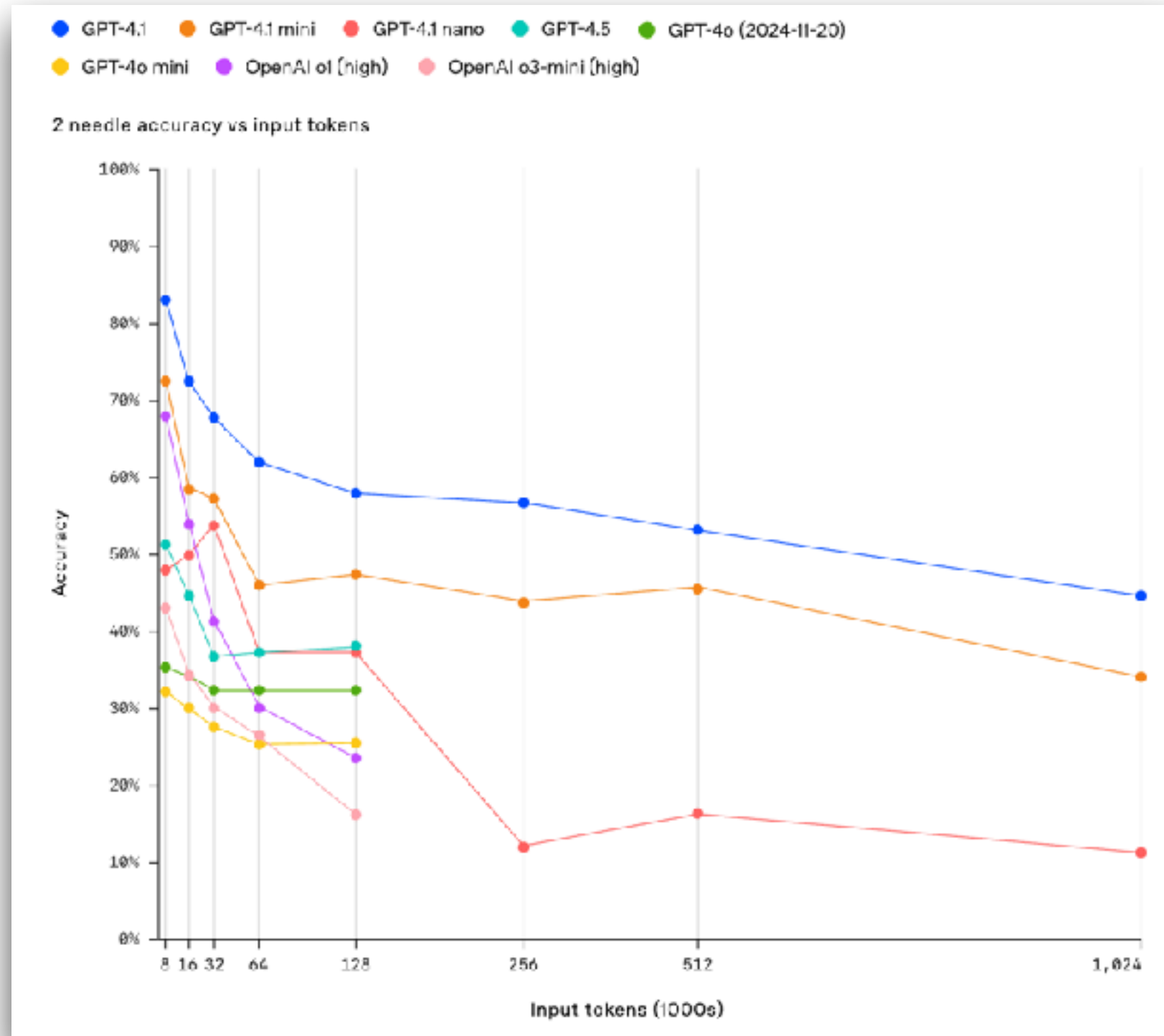
Long Context RAG Performance of LLMs



<https://arxiv.org/abs/2411.03538>



# Long context LLM is better !!



<https://openai.com/index/gpt-4-1/>

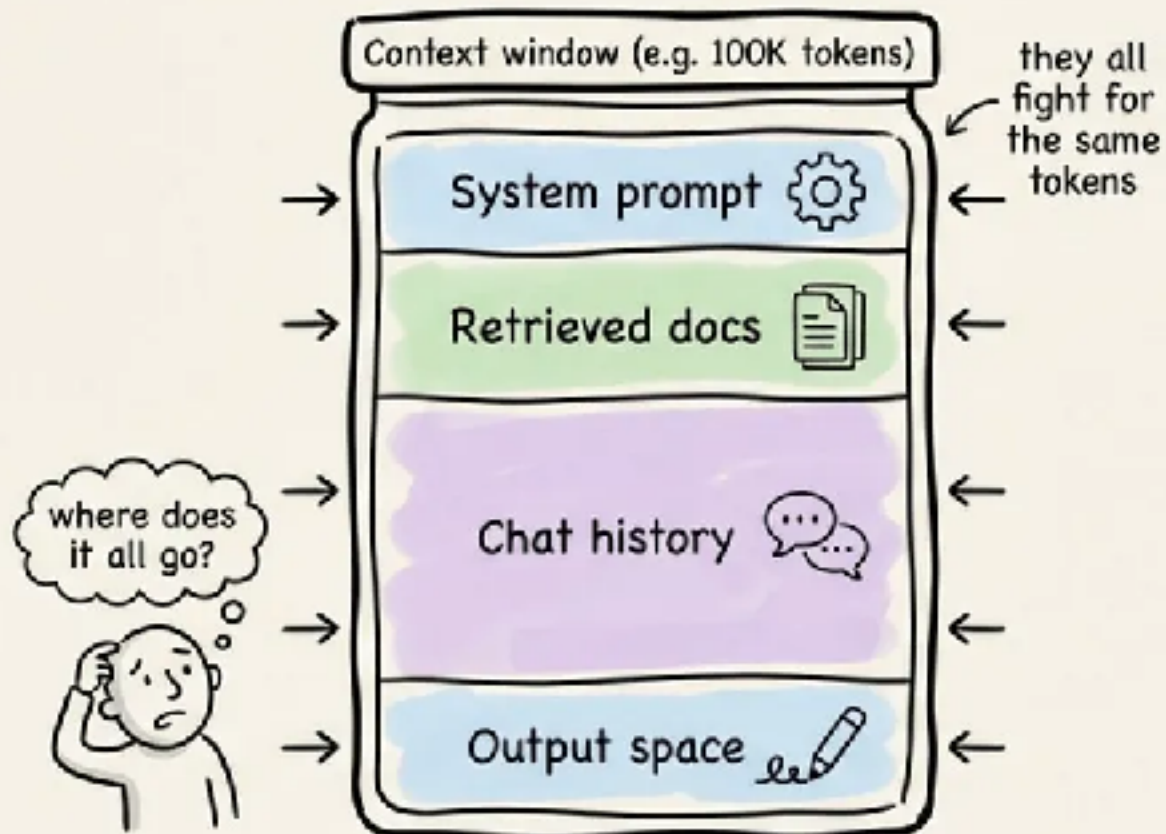


# **Better context == Better result ?**

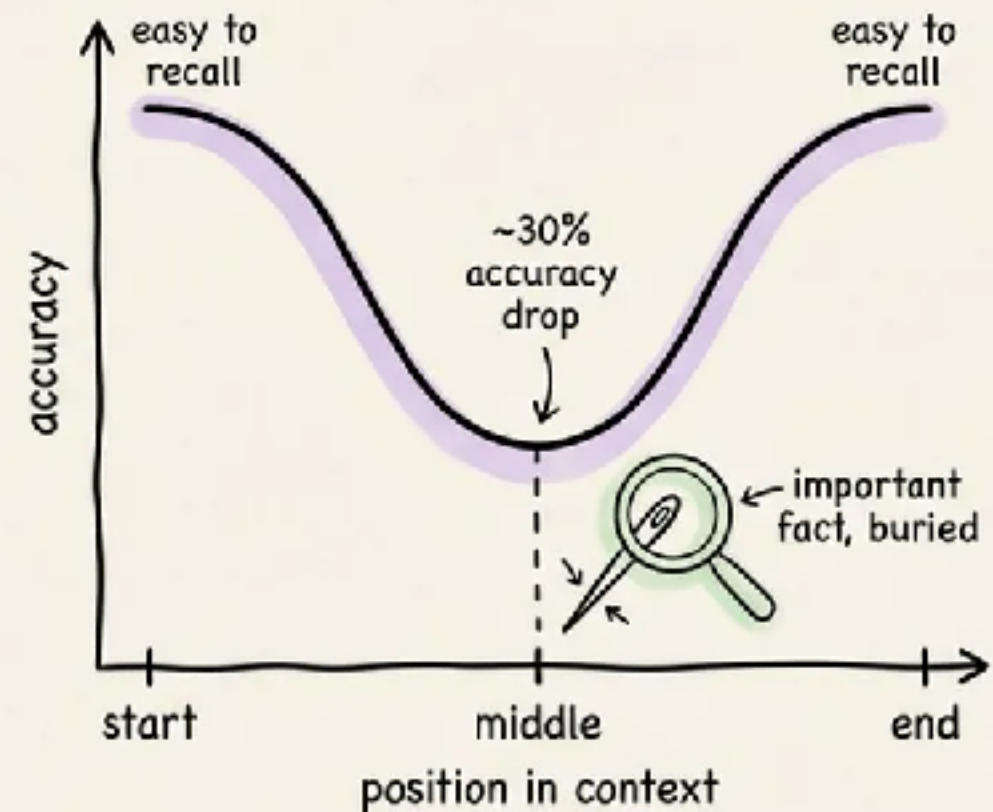


# More tokens isn't more memory.

Context = a shared budget



Lost in the middle



**Bigger windows don't beat structured memory.**

<https://blog.dailydoseofds.com/p/evolution-of-agent-landscape-from>

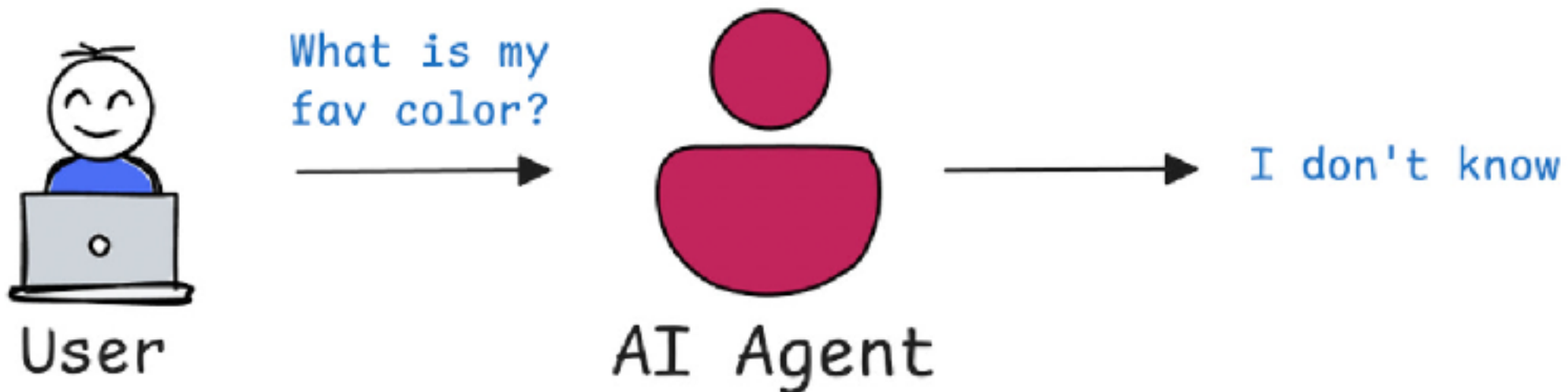
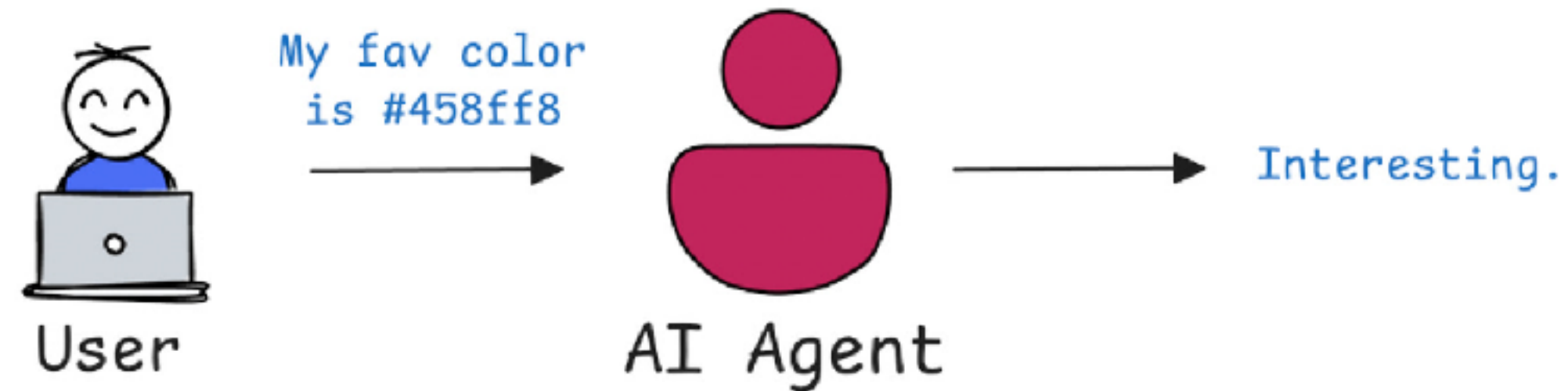


# Agent Memory Memory Bank





# Interaction without memory

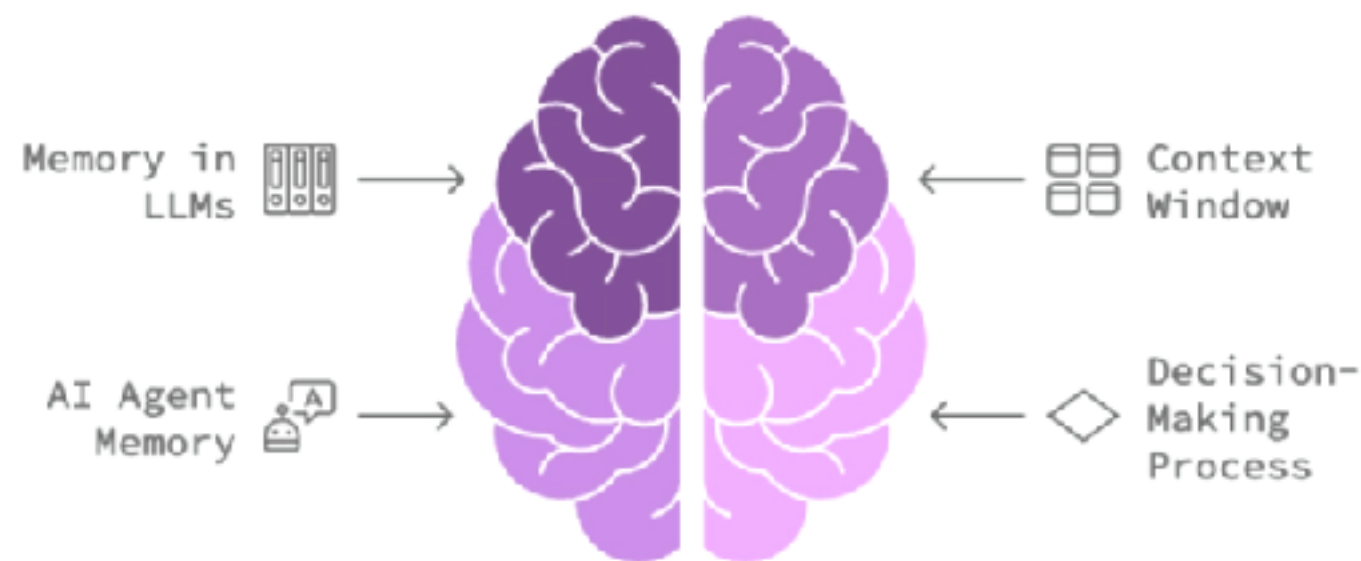


# Why AI Agent memory important ?

Enable personalized and context-aware interactions

Improve decision-making and adaptability

Enhance LLM's performance and response



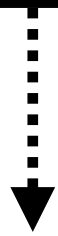
<https://www.falkordb.com/blog/ai-agents-memory-systems/>



# Types of Memory

To make intelligent decisions  
Agents combine different types of memory

Short-term



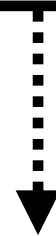
Current goals and steps  
File system (markdown, JSON)  
In-memory database

Episodic



Logs of past interactions  
Success and failure

Semantic

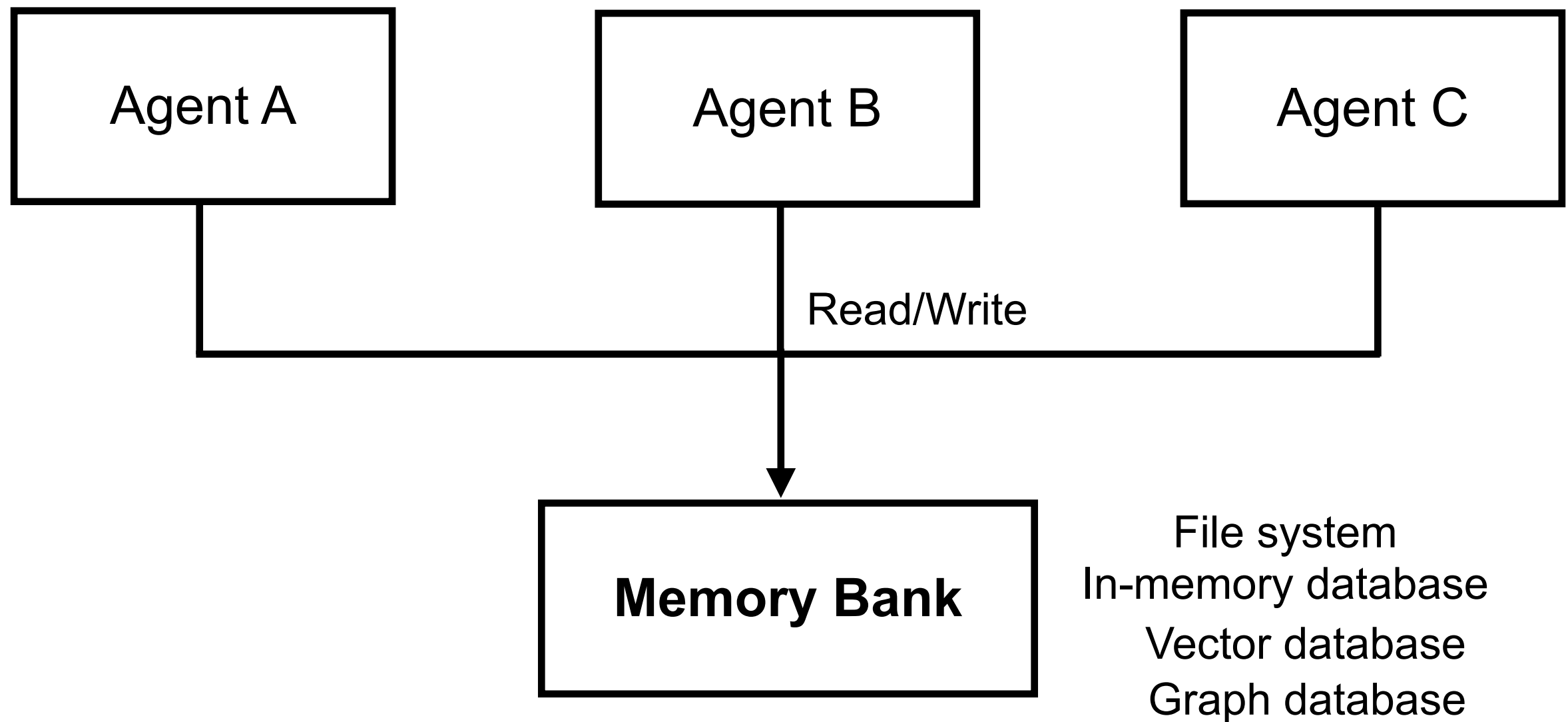


Long-term knowledge  
Fact and strategies  
RAG  
Vector and graph database

<https://bhavishyapandit9.substack.com/p/how-memory-works-in-agentic-ai-a>

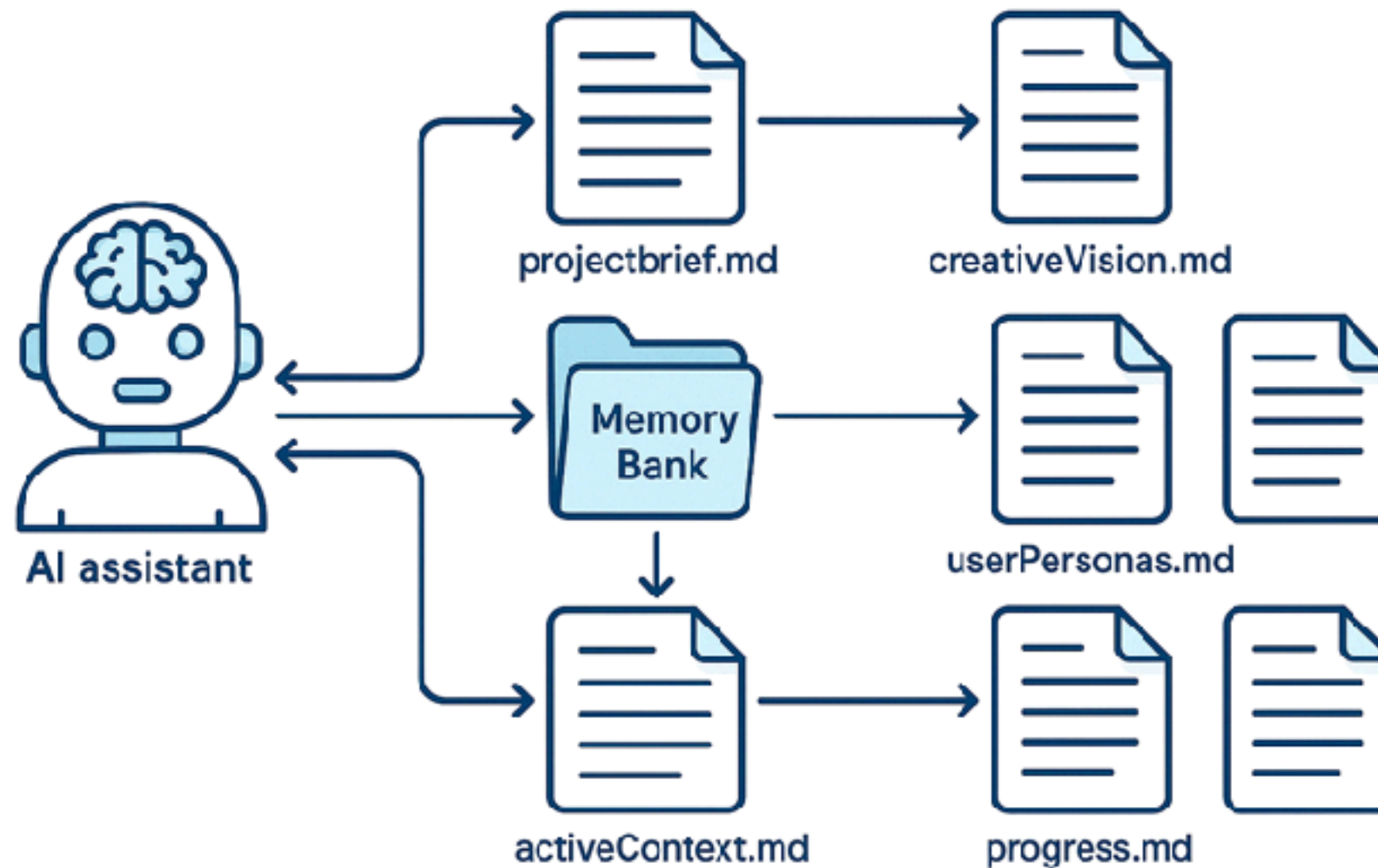


# Shared memory between Agent



# Cline Memory Bank

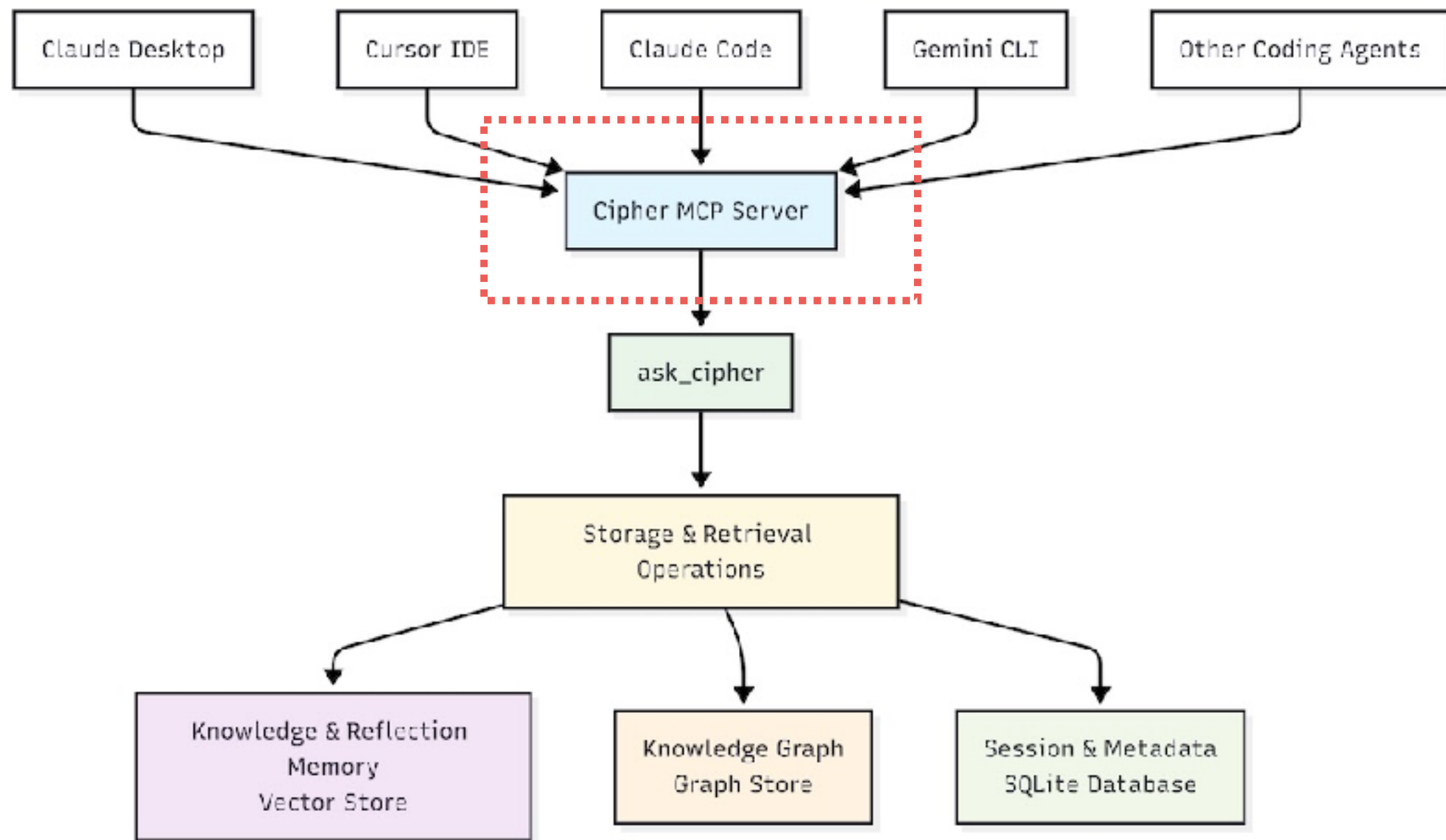
## Working with file system



<https://docs.cline.bot/prompting/cline-memory-bank>



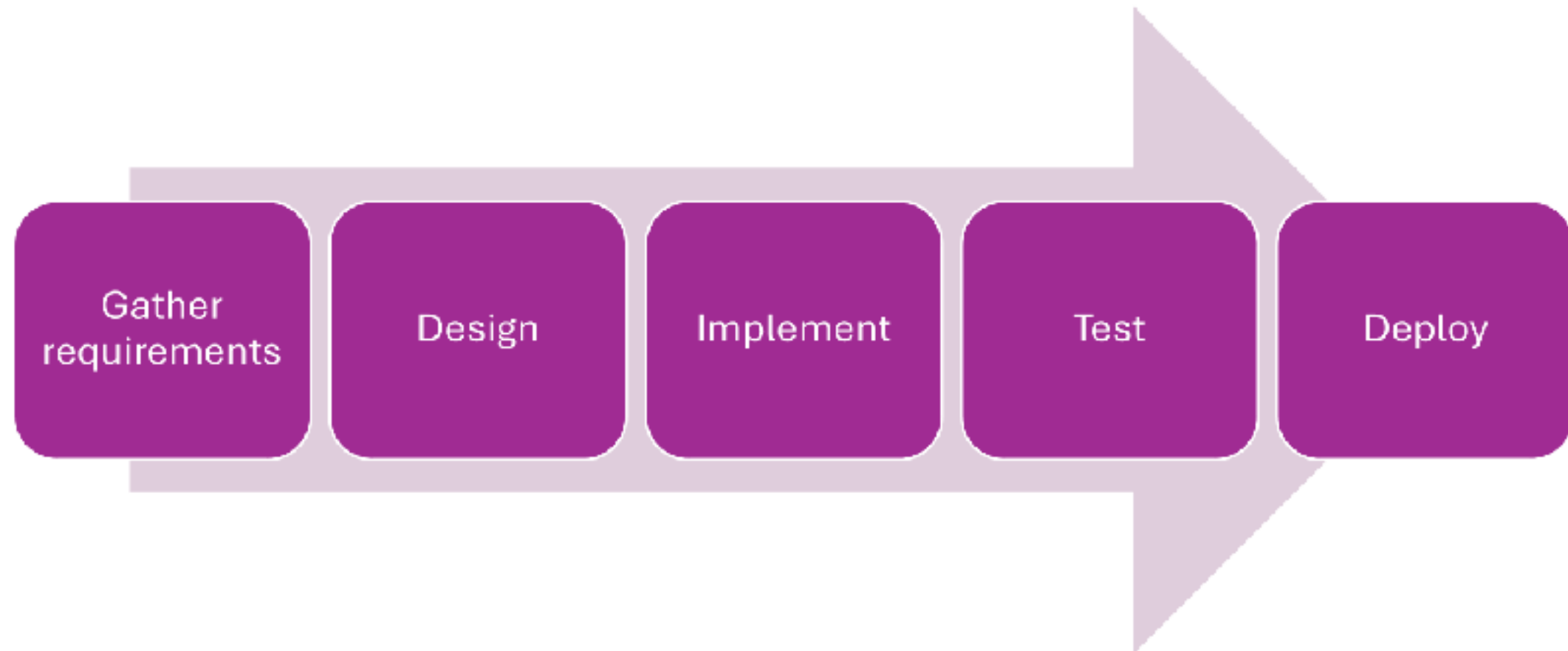
# Memory Bank via MCP



<https://www.somkiat.cc/share-memory-for-ai-ide/>



# Design your AI Agents ?

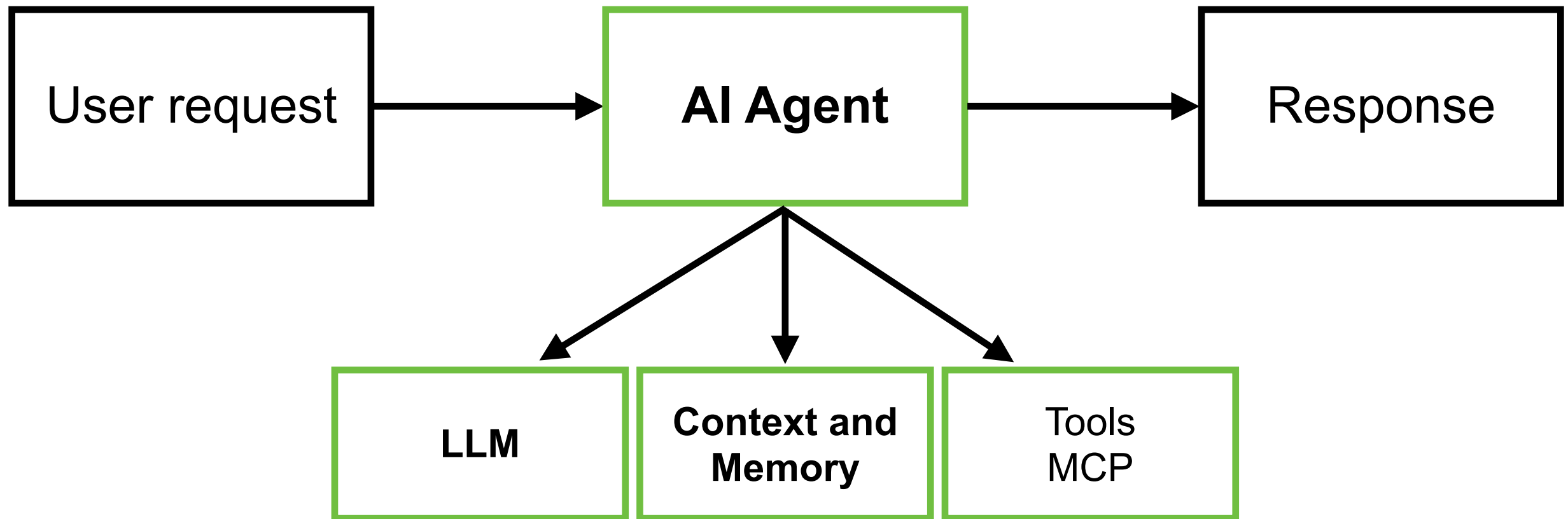


<https://devblogs.microsoft.com/dotnet/introducing-microsoft-agent-framework-preview/>





# Single Agent



# **Sub-agent and Multi-agent !! (Teamwork)**

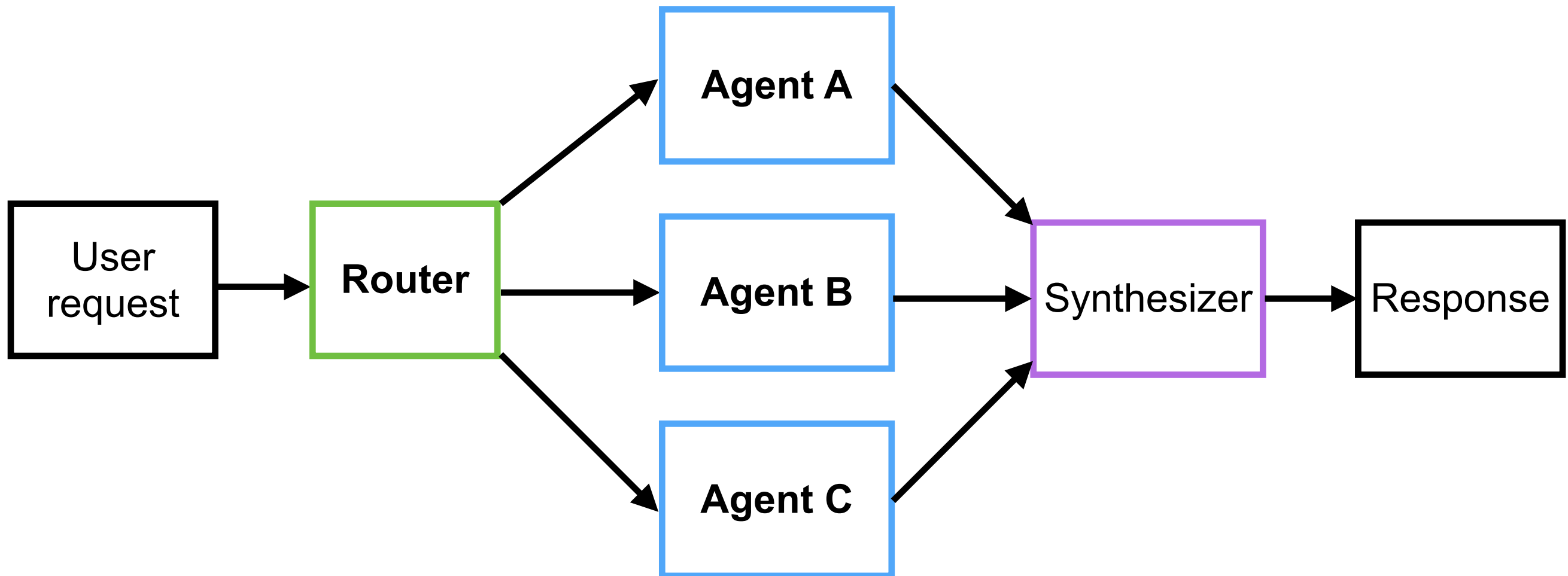
**Reduce size of knowledge !!**

**Single Responsibility Principle**



# Multi AI Agents !!

Router and parallel run



# **Manage context !! In Software development**



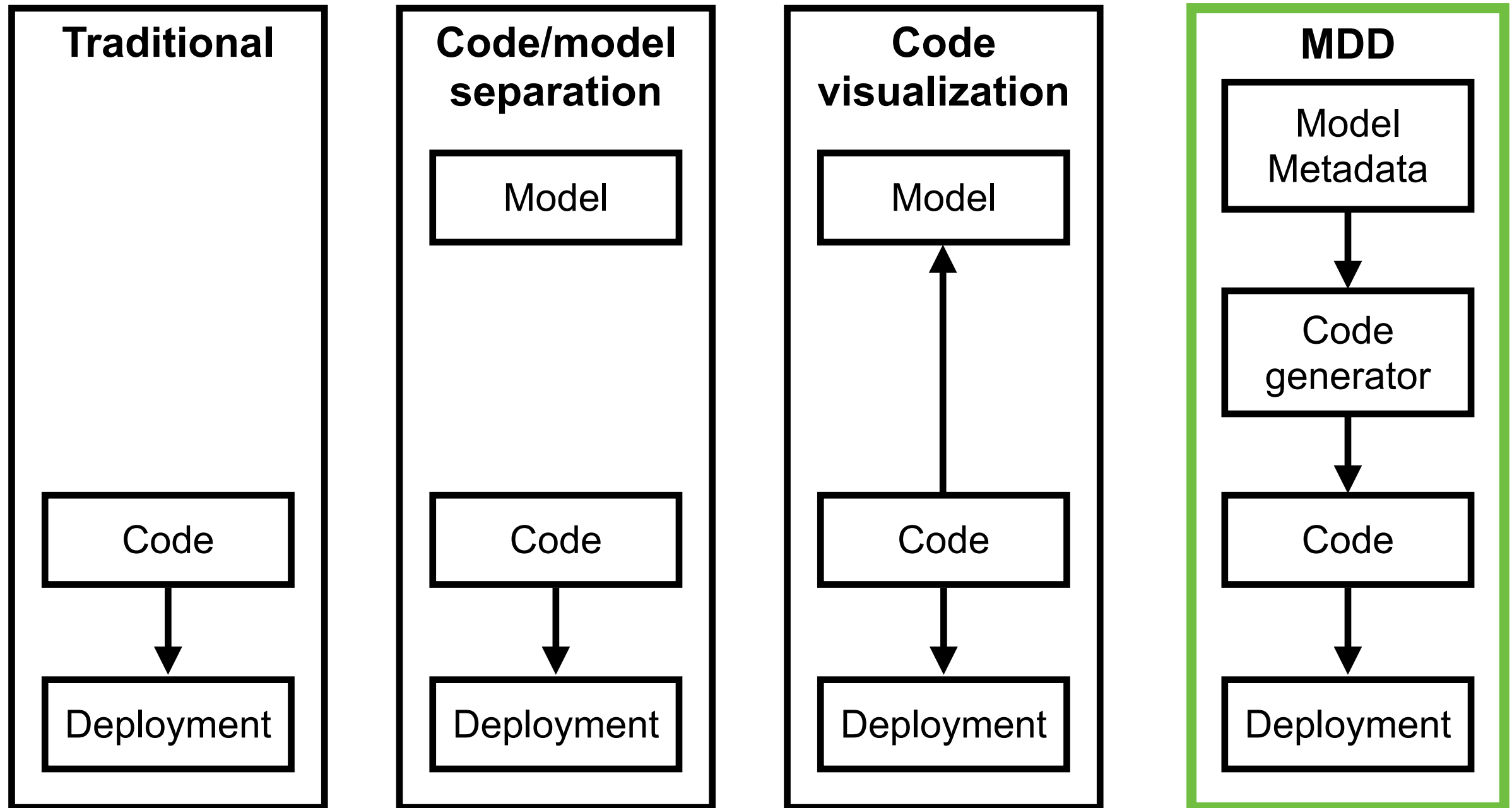
**Specification ?**  
**Architecture ?**  
**Modeling ?**  
**Coding convention ?**  
**Project structure ?**  
**Better practice ?**  
**Technology stack ?**



# Model-Driven Development (MDD)



# Traditional coding to MDD

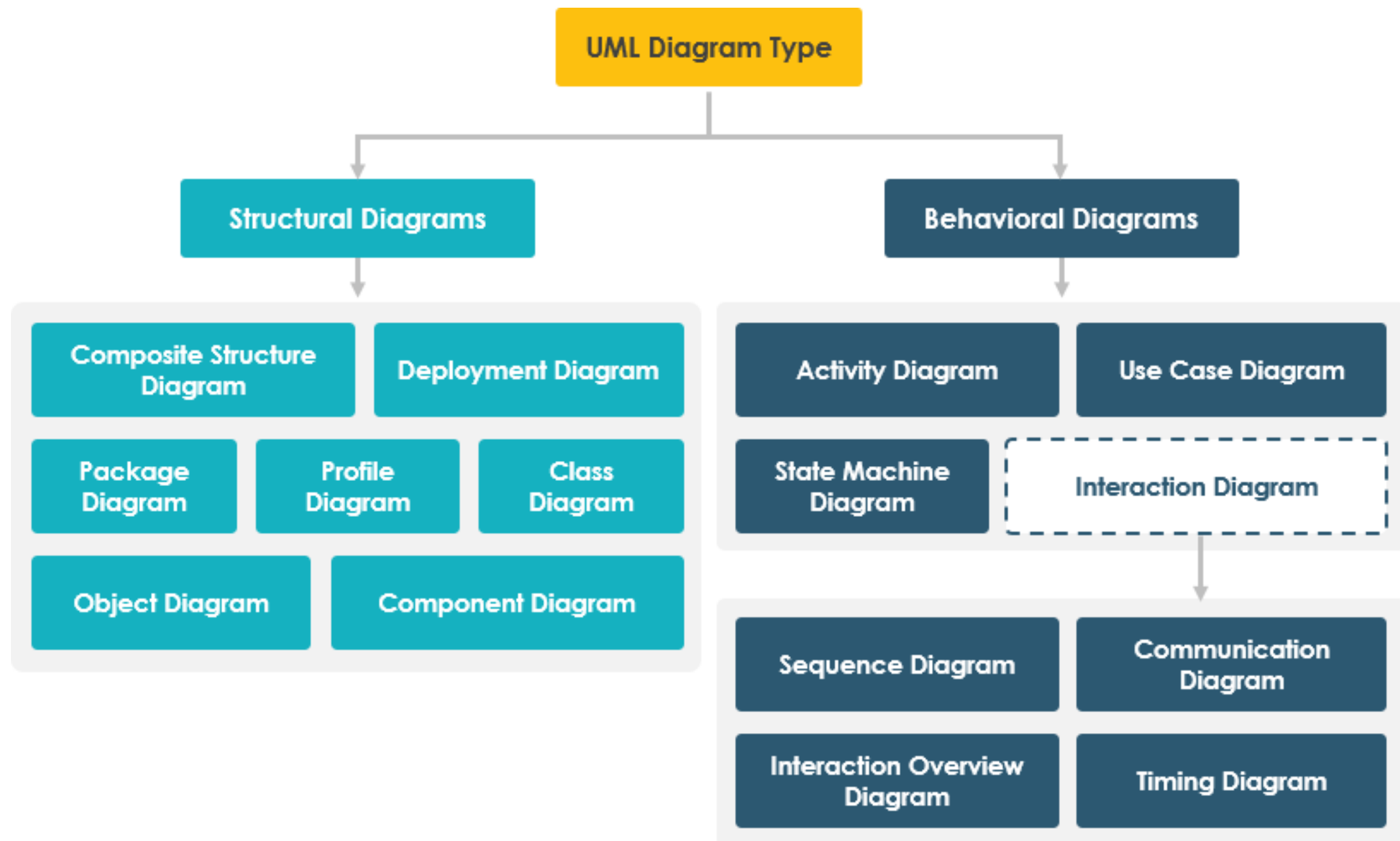


<https://arxiv.org/abs/2410.18489>





# UML Diagrams !!



<https://www.visual-paradigm.com/guide/uml-unified-modeling-language/behavior-vs-structural-diagram/>





# Mermaid Diagram

Diagram as a Code (AI-friendly)

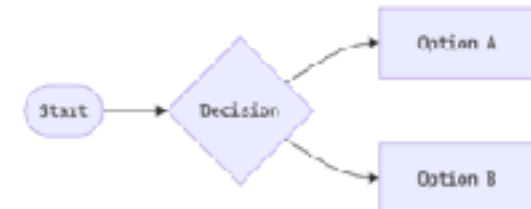
## Mermaid Diagramming and charting tool

JavaScript based diagramming and charting tool that renders Markdown-inspired text definitions to create and modify diagrams dynamically.

Try Editor

Get started

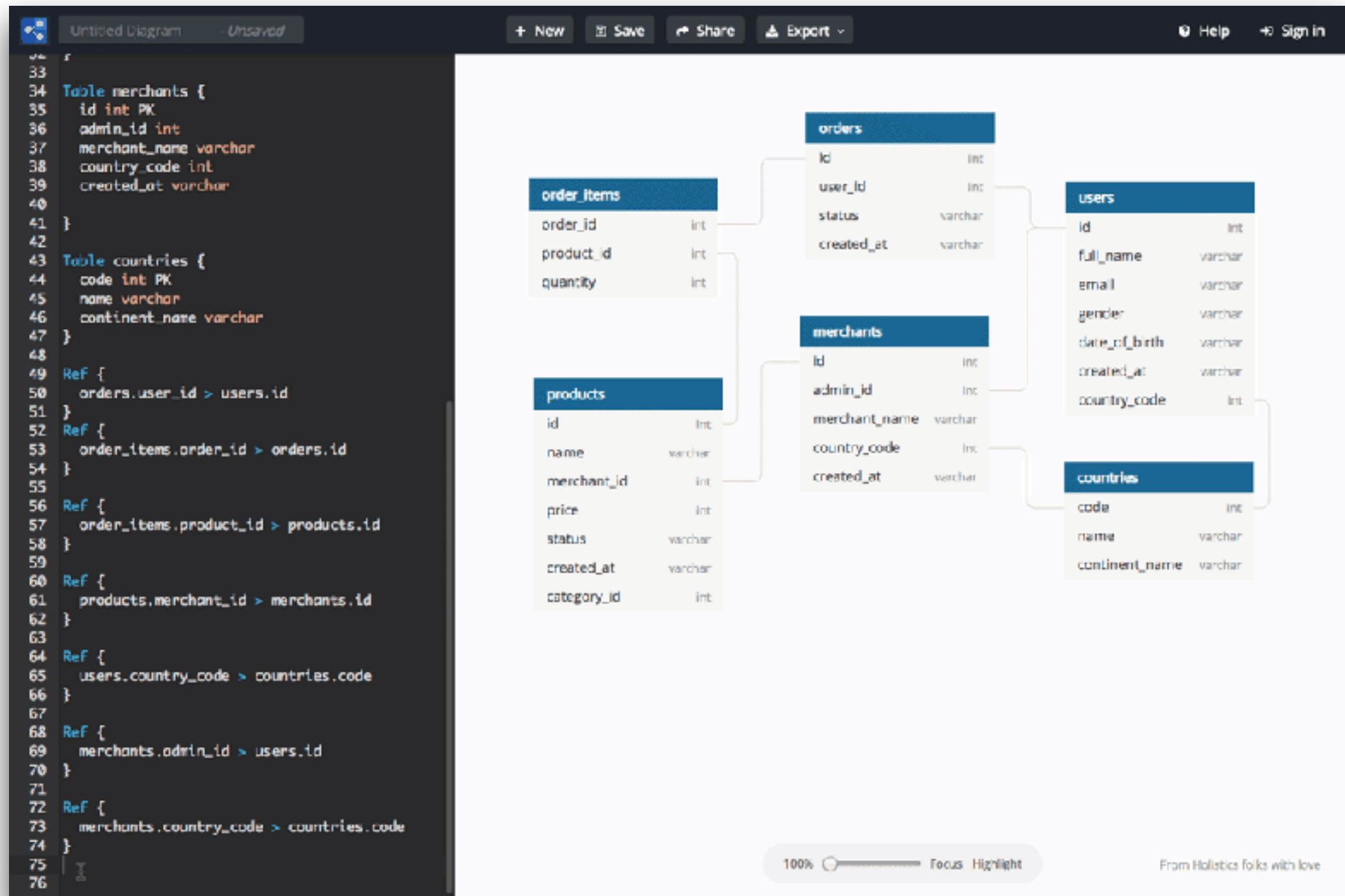
```
1 flowchart LR
2   A(["Start"])
3   A --> B["Decision"]
4   B --> C["Option A"]
5   B --> D["Option B"]
```



<https://mermaid.js.org/>



# ER Diagrams for Database !!



<https://dbml.dbdiagram.io/home/>



# Swagger or OpenAPI

Standardized specification for describe REST APIs

The screenshot displays the Swagger.io homepage. On the left, a hero section titled 'The Future of AI Relies on API Quality' features a green highlight on 'API Quality'. Below the title, text states: 'Swagger enables design, governance, and testing across the full AI-enabled API lifecycle, ensuring quality at every step.' and 'Build APIs ready for humans, LLMs, agents, and continuous innovation.' Two buttons are present: 'Find your tool' and 'Read the docs →'. At the bottom left, it says 'TRUSTED BY'. On the right, a preview of a 'pet' API specification is shown. The title is 'pet Everything about your Pets'. The API endpoints are listed in a table with color-coded rows: POST for image upload, POST for adding a pet, PUT for updating a pet, GET for finding pets by status, GET for finding a pet by ID, POST for updating a pet with form data, and DELETE for deleting a pet. Each row includes the HTTP method, the endpoint path, a brief description, and a lock icon.

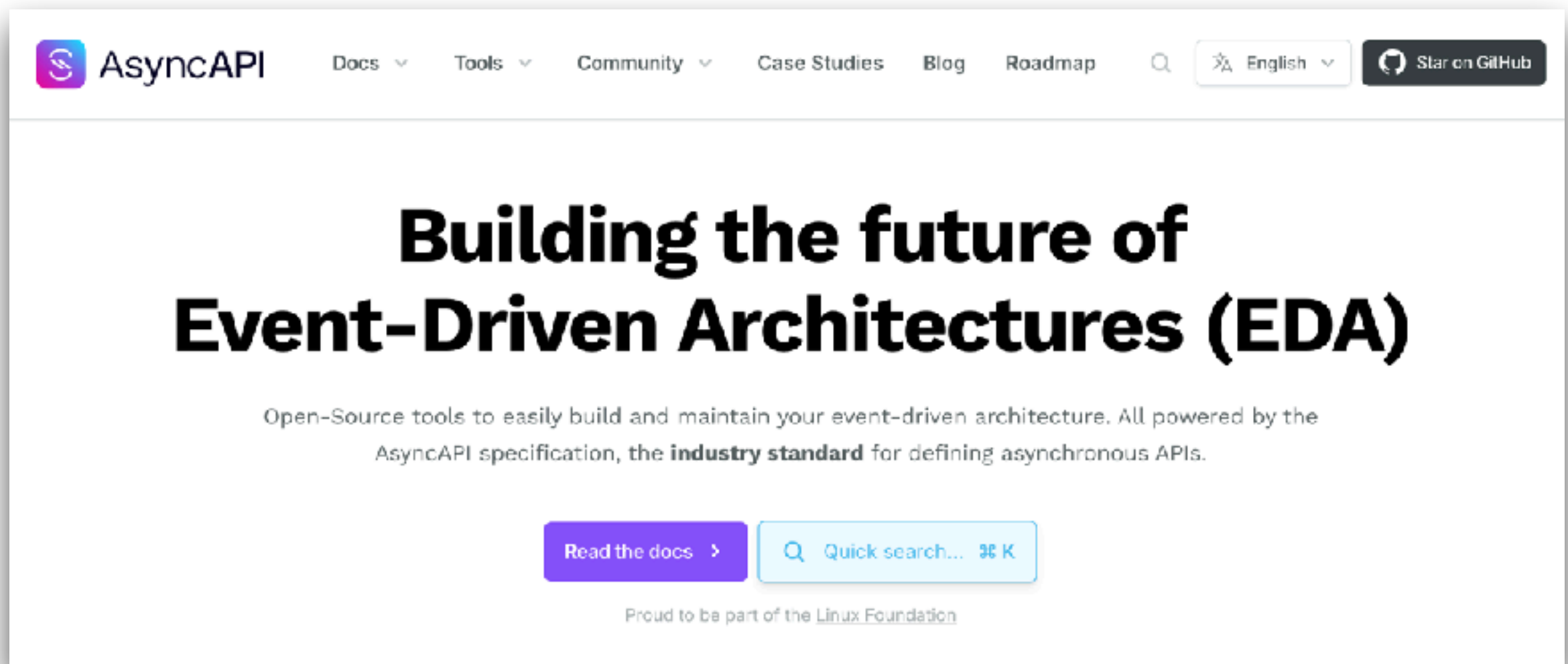
| Method | Endpoint                 | Description                               |
|--------|--------------------------|-------------------------------------------|
| POST   | /pet/{petId}/uploadImage | uploads an image                          |
| POST   | /pet                     | Add a new pet to the store                |
| PUT    | /pet                     | Update an existing pet                    |
| GET    | /pet/findByStatus        | Finds Pets by status                      |
| GET    | /pet/{petId}             | Find pet by ID                            |
| POST   | /pet/{petId}             | Updates a pet in the store with form data |
| DELETE | /pet/{petId}             | Deletes a pet                             |

<https://swagger.io/>



# AsyncAPI

Standardized specification for describe asynchronous and event-driven APIs (Kafka, AMQP, WebSocket)



<https://www.asyncapi.com/>



# JSONSchema

Declarative language used to annotate and validate the structure, constraints and data type of JSON documents

Data validation

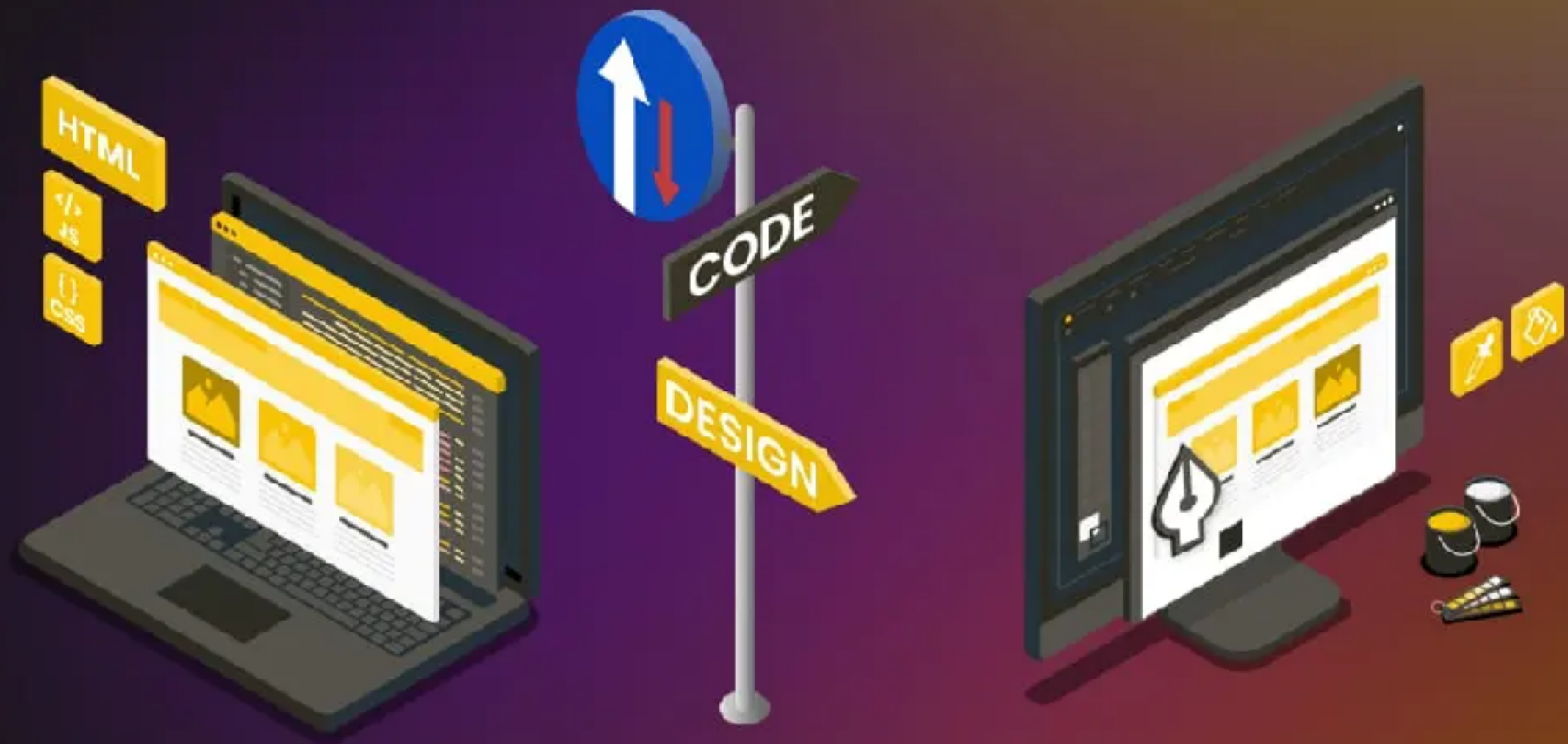
Documentation

Automated  
testing

<https://json-schema.org/>



# Design to Code Code to Design



# Spec-Driven Development (SDD)

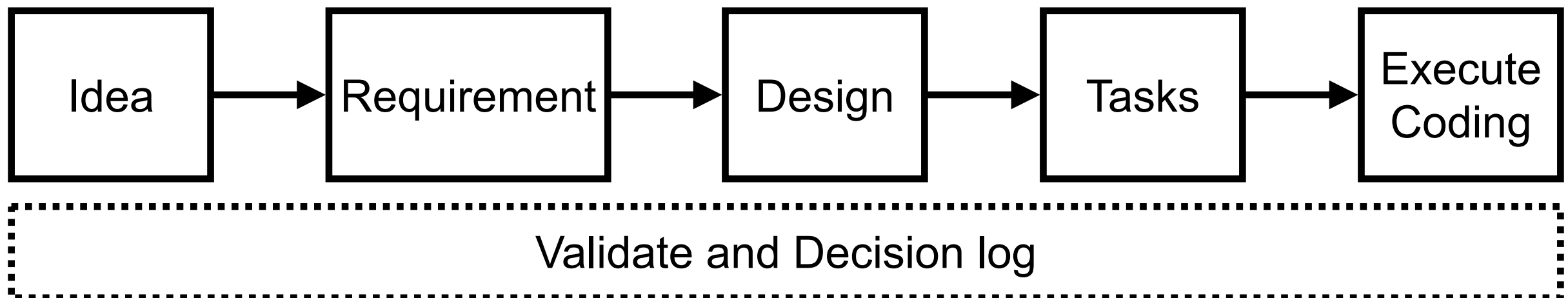




# SDD ?

## Specification or Documentation-First

Modern software development approach where detailed, structure specifications are created before writing code



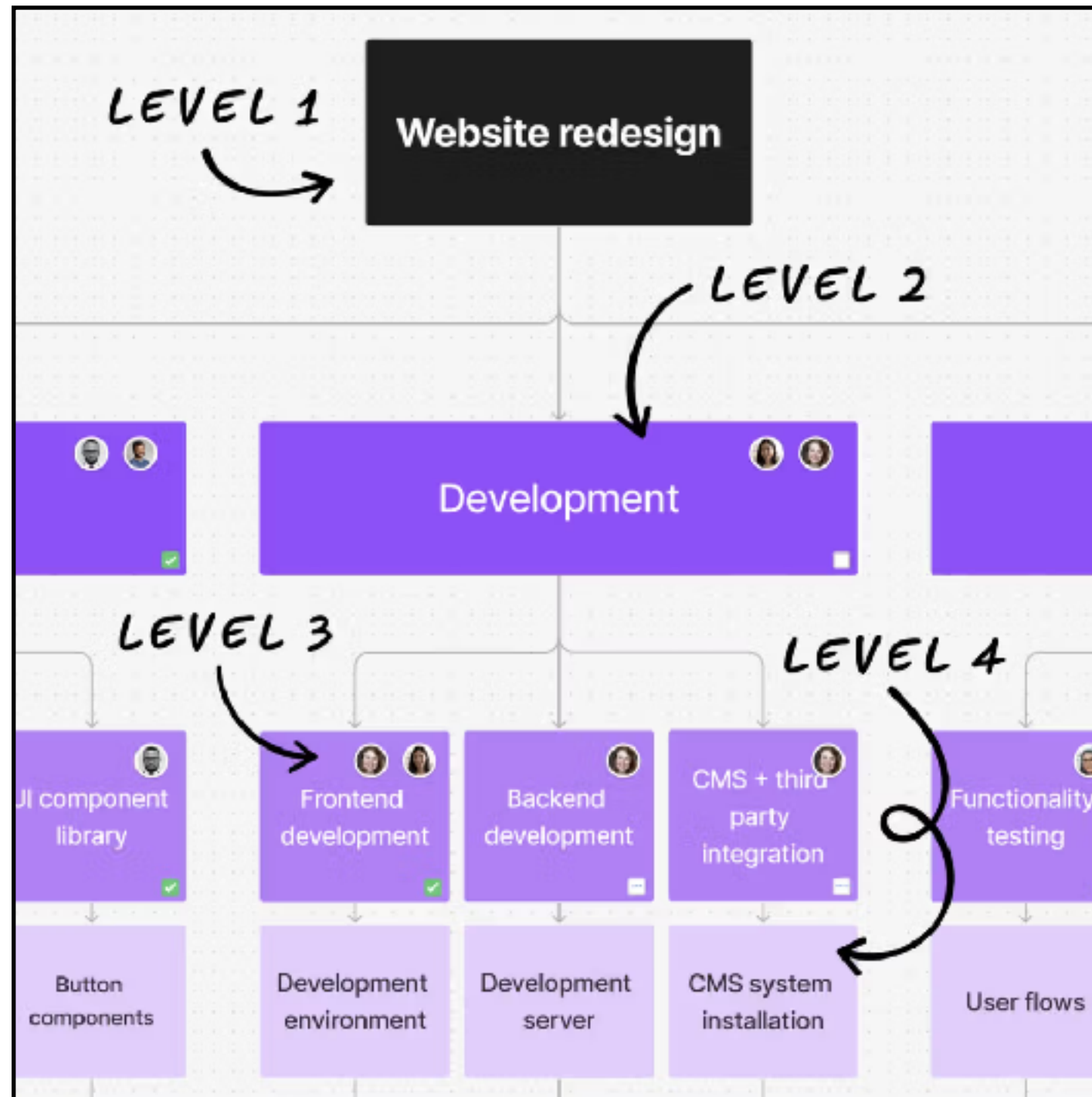


# Types of Specification ?

User requirements  
Design documents  
Interface contracts (UI, API)  
Data schemas  
Validation rules  
Data/business flows  
Security policy and constraints  
Resources and performance constraints



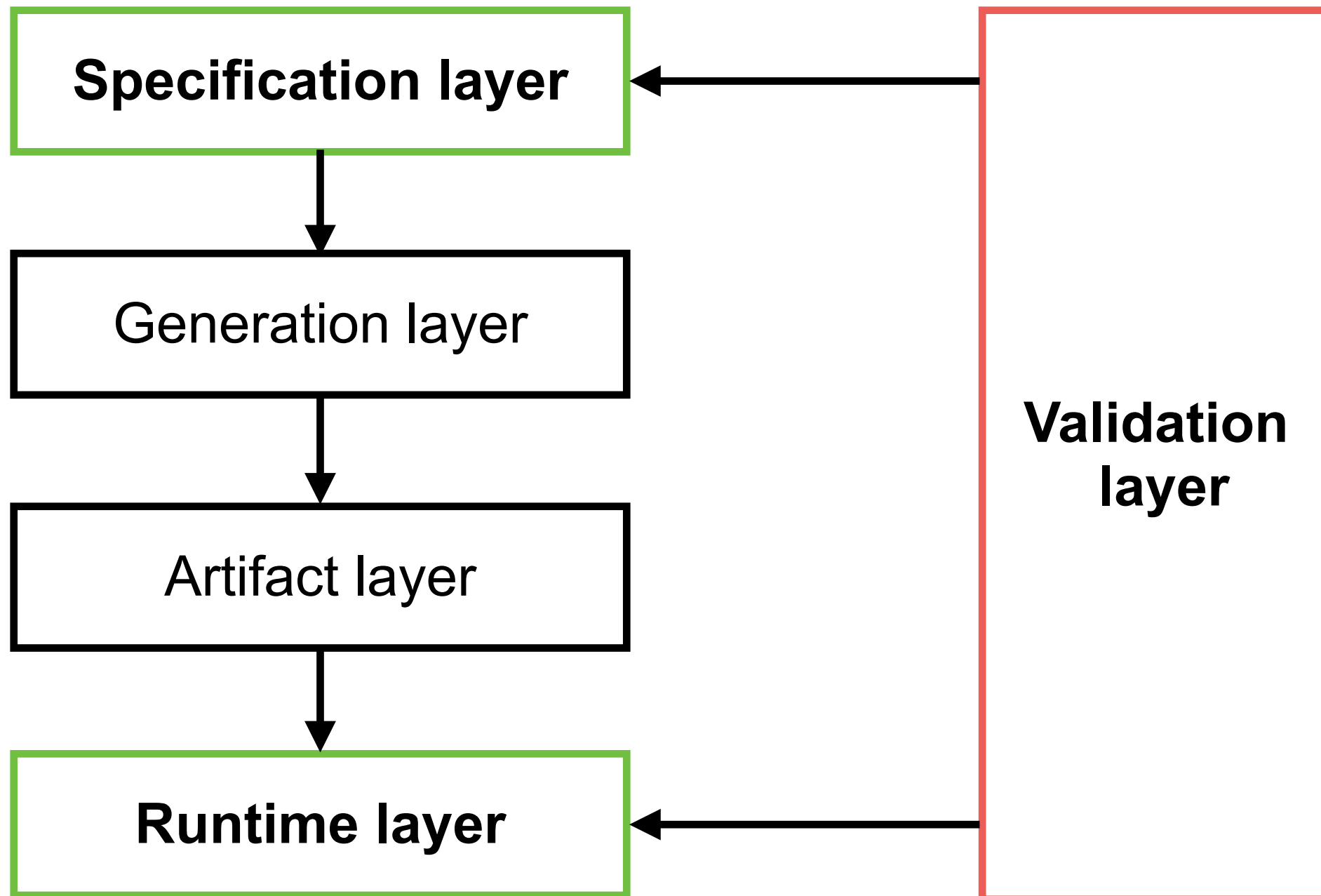
# Work Break Down (WBS)



<https://www.figma.com/resource-library/work-breakdown-structure/>



# SSD Architecture



<https://www.infoq.com/articles/spec-driven-development>



**MDD + SDD == Better result**



# Developer Survival Checklists



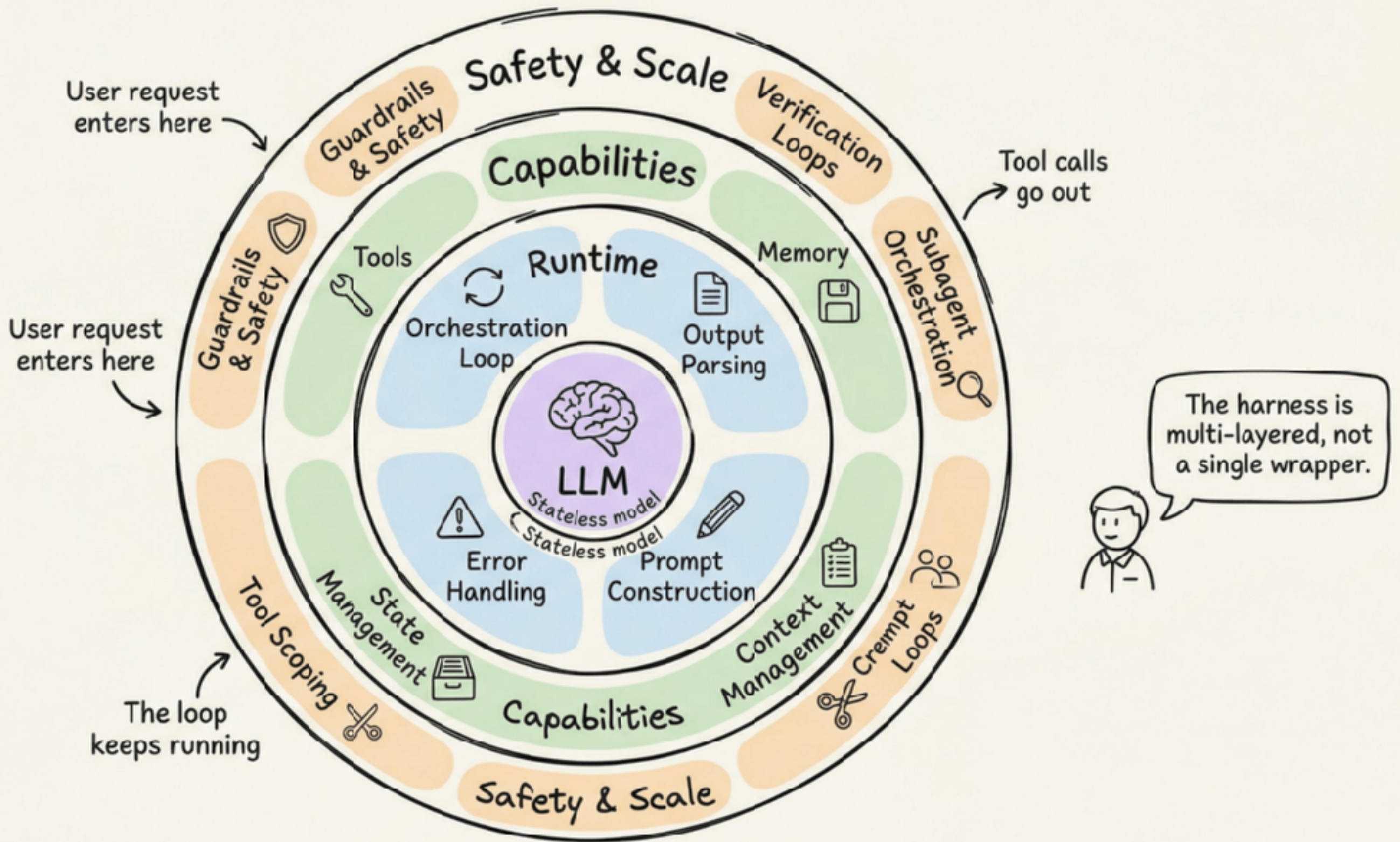
# Developer Survival Checklists

1. Focus on logic
2. Audit AI Output (functional and non-functional)
3. Continuous Learning

**Focus on foundation knowledge  
to effectively work with and guide AI system**







<https://blog.dailydoseofds.com/p/evolution-of-agent-landscape-from>



# Demo time





**Frontend**  
Mobile application

**Backend**  
RESTful API + Database

Write specification

Coding with AI Agent

Run, Test and Verify !!

<https://github.com/up1/demo-sharing-ai-agent-202605>



# Slide and Demo

